

Differentiating $y = f(x)$

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Differentiation (ref d2)

This document contains all 70 questions on *differentiating* $y = f(x)$, each with a fully worked solution and a TikZ diagram: the curve in blue with its **derivative** $f'(x)$ in teal for the pure differentiation questions; the curve with the **tangent** (dashed orange) at the relevant point for gradient, tangent, and stationary-point questions; and additionally the **normal** (dash-dotted purple) for the normal questions.

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Differentiating $y = f(x)$ 01 (d2-001)

Difficulty: Foundation / Marks: 1

Differentiate $y = x^5$.

Worked solution.

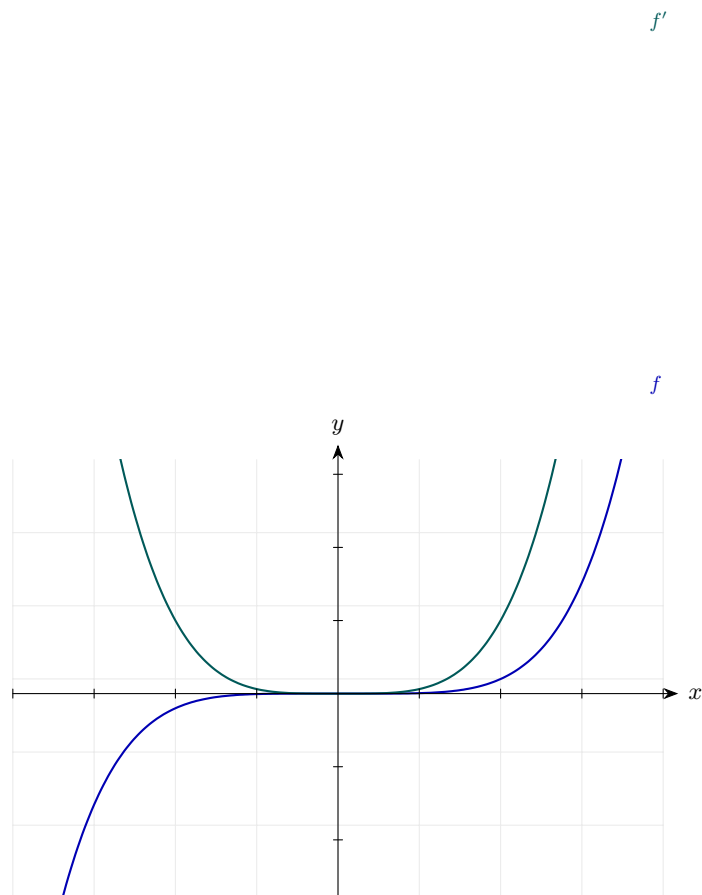
Apply the power rule: multiply by the index, then reduce the index by 1.

$$\frac{dy}{dx} = 5x^{5-1} = 5x^4$$

The rule is: if $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$.

Final answer: $\frac{dy}{dx} = 5x^4$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 02 (d2-002)

Difficulty: Foundation / Marks: 1

Differentiate $y = x^7$.

Worked solution.

Apply the power rule.

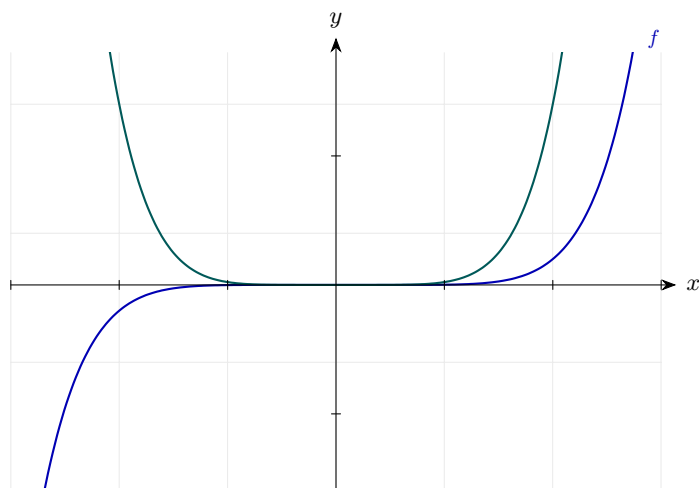
$$\frac{dy}{dx} = 7x^6$$

Multiply the coefficient by the power (here coefficient is 1), then reduce the power by 1.

Final answer: $\frac{dy}{dx} = 7x^6$

Diagram.

f'



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 03 (d2-003)

Difficulty: Foundation / Marks: 1

Differentiate $y = 6x^4$.

Worked solution.

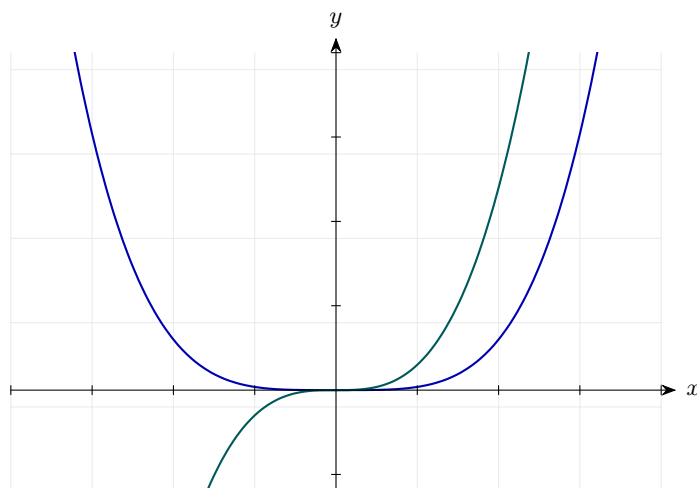
Multiply the coefficient by the power, then reduce the power by 1.

$$\frac{dy}{dx} = 6 \times 4x^3 = 24x^3$$

When there is a number in front, multiply it by the index.

Final answer: $\frac{dy}{dx} = 24x^3$

Diagram.

f' f 

Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 04 (d2-004)

Difficulty: Foundation / Marks: 2

Differentiate $y = \sqrt{x}$.

Worked solution.

Step 1: Rewrite the square root as a fractional power.

$$y = x^{1/2}$$

$\sqrt{x} = x^{1/2}$. Always convert roots to fractional powers before differentiating.

Step 2: Apply the power rule.

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2}$$

Multiply by $\frac{1}{2}$ and reduce the index to $\frac{1}{2} - 1 = -\frac{1}{2}$.

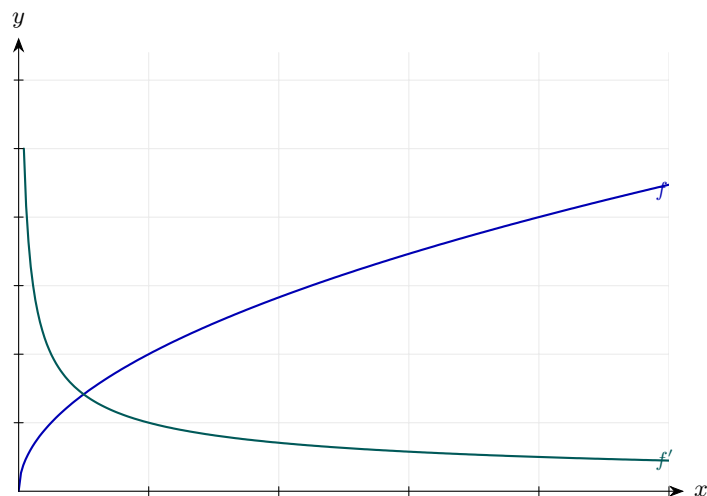
Step 3: Rewrite using a surd if required.

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

$$x^{-1/2} = \frac{1}{\sqrt{x}}.$$

Final answer: $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 05 (d2-005)

Difficulty: Foundation / Marks: 2

Differentiate $y = \frac{1}{x^3}$.

Worked solution.

Step 1: Rewrite as a negative power of x .

$$y = x^{-3}$$

$\frac{1}{x^3} = x^{-3}$. Convert before differentiating.

Step 2: Apply the power rule.

$$\frac{dy}{dx} = -3x^{-4}$$

Multiply by -3 and reduce the index: $-3 - 1 = -4$.

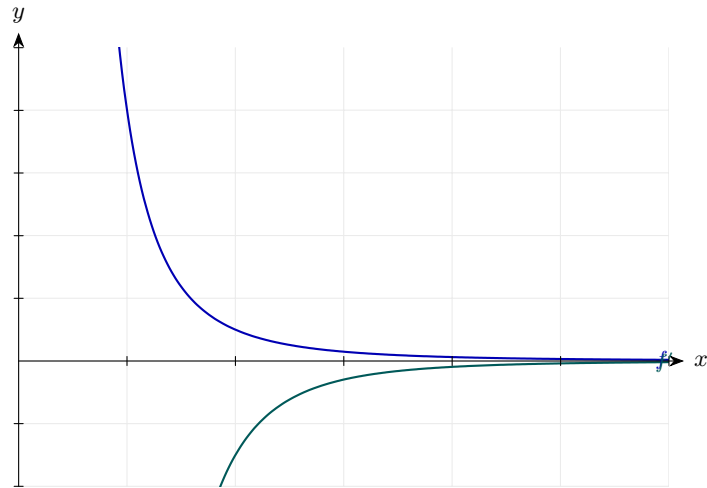
Step 3: Write in fraction form.

$$\frac{dy}{dx} = -\frac{3}{x^4}$$

$$x^{-4} = \frac{1}{x^4}.$$

Final answer: $\frac{dy}{dx} = -\frac{3}{x^4}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 06 (d2-006)

Difficulty: Foundation / Marks: 3

Differentiate $y = 5x^3 - 4x^2 + 7x - 2$.

Worked solution.

Step 1: Differentiate each term separately.

$$\frac{dy}{dx} = 3(5x^2) - 2(4x) + 7 - 0$$

The derivative of a sum is the sum of the derivatives. The constant -2 differentiates to 0.

Step 2: Simplify.

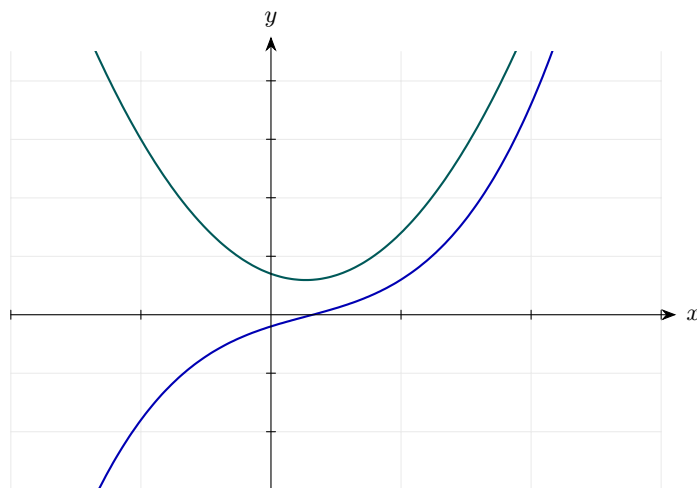
$$\frac{dy}{dx} = 15x^2 - 8x + 7$$

Collect the simplified terms.

Final answer: $\frac{dy}{dx} = 15x^2 - 8x + 7$

Diagram.

f'
 f



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 07 (d2-007)

Difficulty: Foundation / Marks: 2

Differentiate $f(x) = 2x^4 - 9x^2 + 1$.

Worked solution.

Differentiate each term.

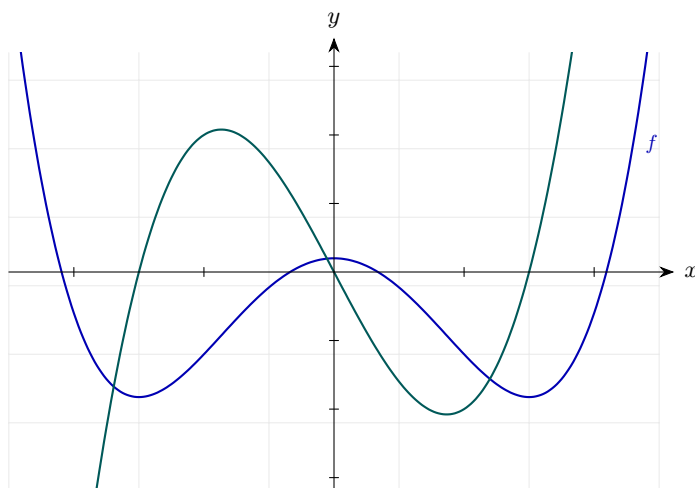
$$f'(x) = 8x^3 - 18x$$

The derivative of $2x^4$ is $8x^3$, of $-9x^2$ is $-18x$, and of the constant 1 is 0.

Final answer: $f'(x) = 8x^3 - 18x$

Diagram.

f'



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 08 (d2-008)

Difficulty: Foundation / Marks: 2

Find $\frac{dy}{dx}$ when $y = 3x^{-2} + 5x$.

Worked solution.

Step 1: Differentiate each term using the power rule.

$$\frac{dy}{dx} = 3(-2)x^{-3} + 5(1)x^0$$

$x^0 = 1$, so the derivative of $5x$ is simply 5.

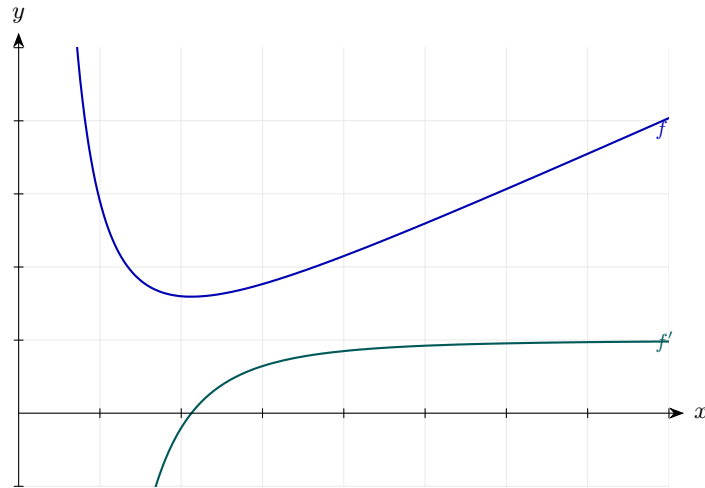
Step 2: Simplify.

$$\frac{dy}{dx} = -6x^{-3} + 5$$

Or equivalently $-\frac{6}{x^3} + 5$.

Final answer: $\frac{dy}{dx} = -\frac{6}{x^3} + 5$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 09 (d2-009)

Difficulty: Foundation / Marks: 3

Expand and differentiate $y = x(x - 3)^2$.

Worked solution.

Step 1: Expand $(x - 3)^2$ first.

$$(x - 3)^2 = x^2 - 6x + 9$$

Square the bracket before multiplying.

Step 2: Multiply by x .

$$y = x(x^2 - 6x + 9) = x^3 - 6x^2 + 9x$$

Distribute the x across every term.

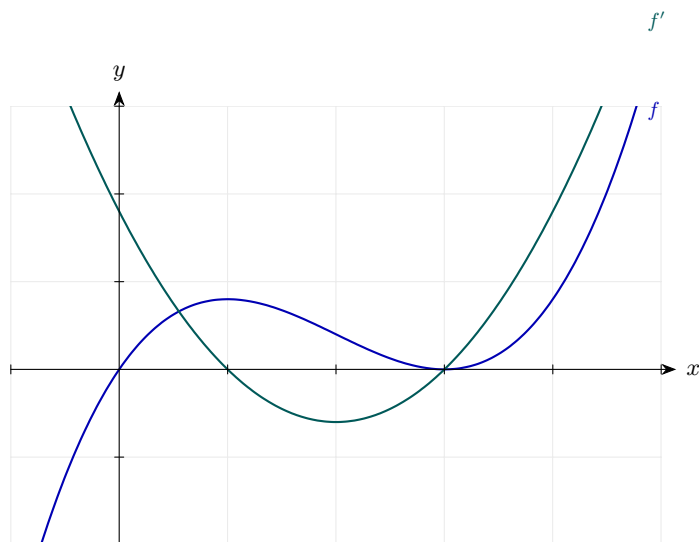
Step 3: Differentiate term by term.

$$\frac{dy}{dx} = 3x^2 - 12x + 9$$

Apply the power rule to each term.

Final answer: $\frac{dy}{dx} = 3x^2 - 12x + 9$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 10 (d2-010)

Difficulty: Foundation / Marks: 3

Expand and differentiate $y = (2x + 1)(x - 4)$.

Worked solution.

Step 1: Expand the brackets.

$$y = 2x^2 - 8x + x - 4 = 2x^2 - 7x - 4$$

FOIL: $2x \cdot x = 2x^2$, $2x \cdot (-4) = -8x$, $1 \cdot x = x$, $1 \cdot (-4) = -4$.

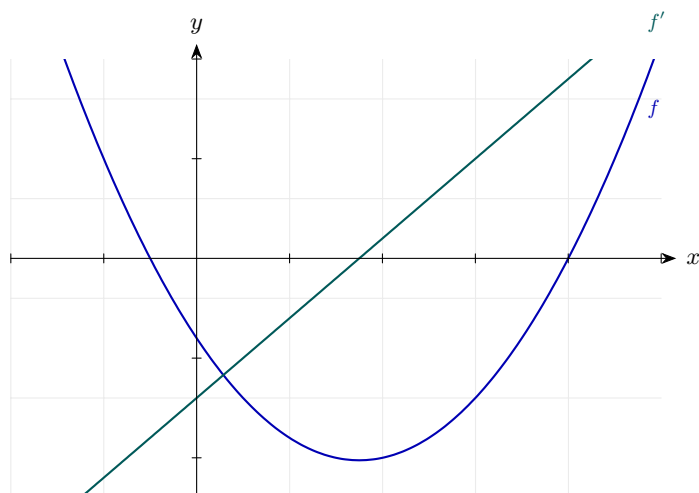
Step 2: Differentiate.

$$\frac{dy}{dx} = 4x - 7$$

Apply the power rule to $2x^2$ and $-7x$; the constant -4 disappears.

Final answer: $\frac{dy}{dx} = 4x - 7$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 11 (d2-011)

Difficulty: Foundation / Marks: 3

Find $f'(x)$ for $f(x) = \frac{x^3 + 4x}{x}$.

Worked solution.

Step 1: Split the fraction by dividing each term in the numerator by x .

$$f(x) = \frac{x^3}{x} + \frac{4x}{x} = x^2 + 4$$

Simplify before differentiating — never try to differentiate a fraction directly in this form.

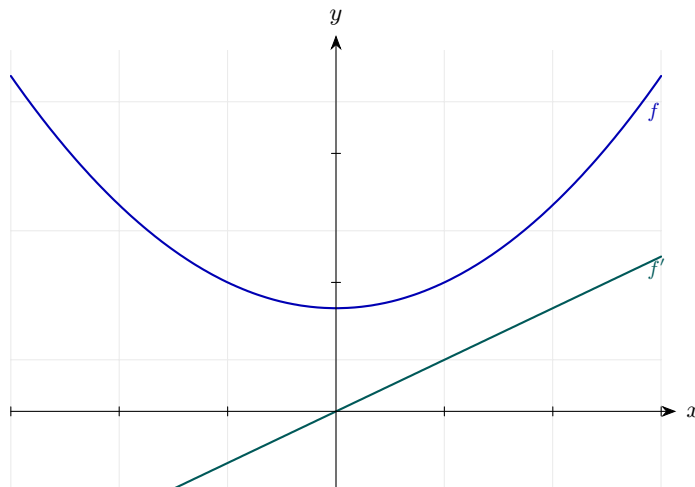
Step 2: Differentiate.

$$f'(x) = 2x$$

The constant 4 differentiates to 0.

Final answer: $f'(x) = 2x$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 12 (d2-012)

Difficulty: Foundation / Marks: 3

Simplify and differentiate $y = \frac{x^4 - 3x^2}{x^2}$.

Worked solution.

Step 1: Divide each term in the numerator by x^2 .

$$y = \frac{x^4}{x^2} - \frac{3x^2}{x^2} = x^2 - 3$$

This gives a simple polynomial to differentiate.

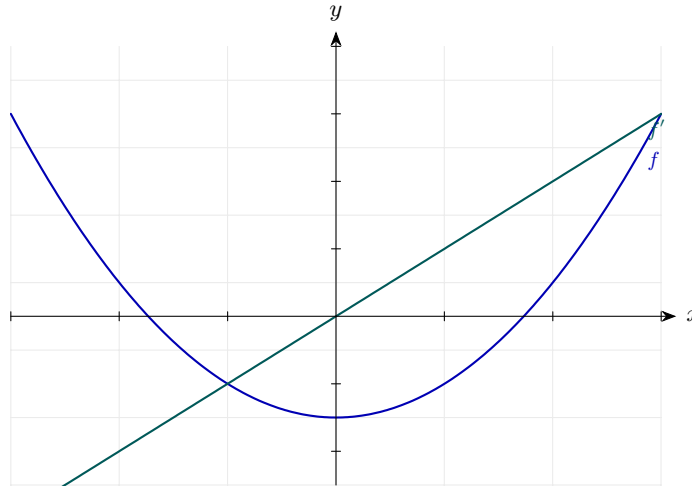
Step 2: Differentiate.

$$\frac{dy}{dx} = 2x$$

The constant -3 differentiates to 0.

Final answer: $\frac{dy}{dx} = 2x$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 13 (d2-013)

Difficulty: Foundation / Marks: 3

Differentiate $y = 4\sqrt{x} - \frac{2}{x}$.

Worked solution.

Step 1: Rewrite using index notation.

$$y = 4x^{1/2} - 2x^{-1}$$

$$\sqrt{x} = x^{1/2} \text{ and } \frac{1}{x} = x^{-1}.$$

Step 2: Differentiate each term.

$$\frac{dy}{dx} = 4 \cdot \frac{1}{2}x^{-1/2} - 2(-1)x^{-2} = 2x^{-1/2} + 2x^{-2}$$

Apply the power rule to each term separately.

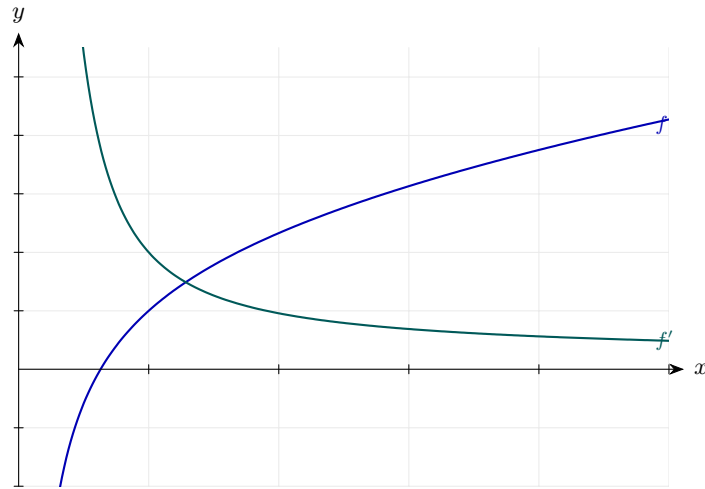
Step 3: Rewrite in standard form.

$$\frac{dy}{dx} = \frac{2}{\sqrt{x}} + \frac{2}{x^2}$$

Convert negative/fractional indices back to fractions or surds.

Final answer: $\frac{dy}{dx} = \frac{2}{\sqrt{x}} + \frac{2}{x^2}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 14 (d2-014)

Difficulty: Foundation / Marks: 3

Find the gradient of $y = 3x^2 - 5x + 1$ at $x = 2$.

Worked solution.

Step 1: Differentiate y .

$$\frac{dy}{dx} = 6x - 5$$

Power rule applied to each term.

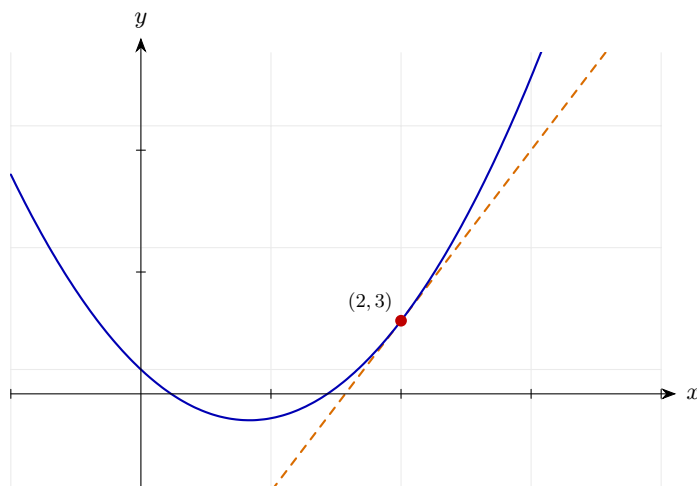
Step 2: Substitute $x = 2$.

$$\left. \frac{dy}{dx} \right|_{x=2} = 6(2) - 5 = 12 - 5 = 7$$

Substitute the given x -value into the derivative.

Final answer: Gradient = 7

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 15 (d2-015)

Difficulty: Foundation / Marks: 3

Find the gradient of the curve $y = 2x^3 - x$ at the point $(1, 1)$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 6x^2 - 1$$

Apply the power rule to each term.

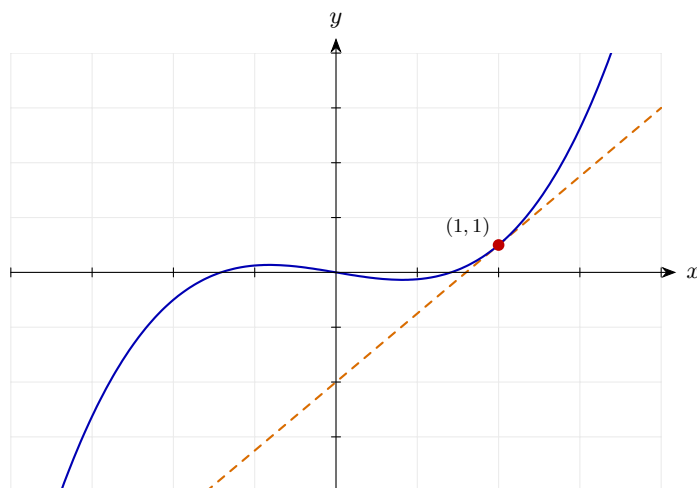
Step 2: Substitute $x = 1$.

$$\frac{dy}{dx} = 6(1)^2 - 1 = 5$$

The gradient at $(1, 1)$ is the value of the derivative when $x = 1$.

Final answer: Gradient = 5

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 16 (d2-016)

Difficulty: Foundation / Marks: 3

Find the gradient of $f(x) = 2\sqrt{x}$ at the point $(9, 6)$.

Worked solution.

Step 1: Rewrite $f(x)$ using index notation.

$$f(x) = 2x^{1/2}$$

Convert the square root to a fractional power.

Step 2: Differentiate.

$$f'(x) = 2 \cdot \frac{1}{2}x^{-1/2} = x^{-1/2} = \frac{1}{\sqrt{x}}$$

Apply the power rule and simplify.

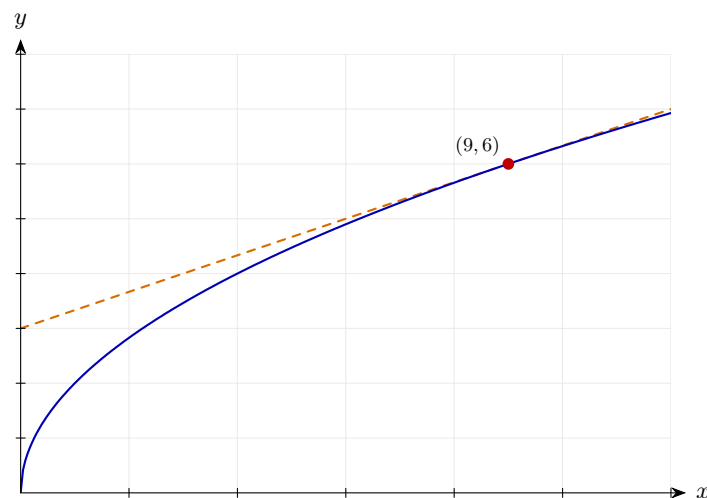
Step 3: Substitute $x = 9$.

$$f'(9) = \frac{1}{\sqrt{9}} = \frac{1}{3}$$

$$\sqrt{9} = 3.$$

Final answer: Gradient = $\frac{1}{3}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 17 (d2-017)

Difficulty: Foundation / Marks: 3

Find the gradient of $y = x^4 - 3x^2 + 2$ when $x = -1$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 4x^3 - 6x$$

Term-by-term differentiation.

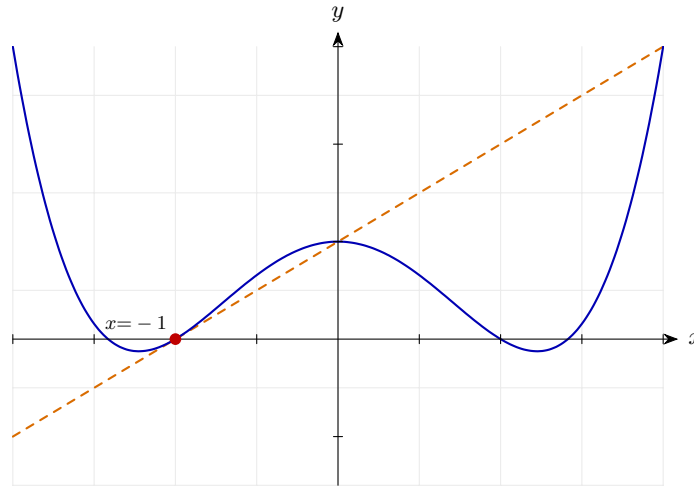
Step 2: Substitute $x = -1$.

$$\frac{dy}{dx} = 4(-1)^3 - 6(-1) = -4 + 6 = 2$$

Be careful with negative values: $(-1)^3 = -1$.

Final answer: Gradient = 2

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 18 (d2-018)

Difficulty: Foundation / Marks: 4

Find the coordinates of the point on $y = x^2 - 6x + 5$ where the gradient is 0.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 2x - 6$$

The gradient function gives the slope at any point.

Step 2: Set the derivative equal to 0.

$$2x - 6 = 0 \implies x = 3$$

A gradient of 0 means the curve is neither rising nor falling at that point.

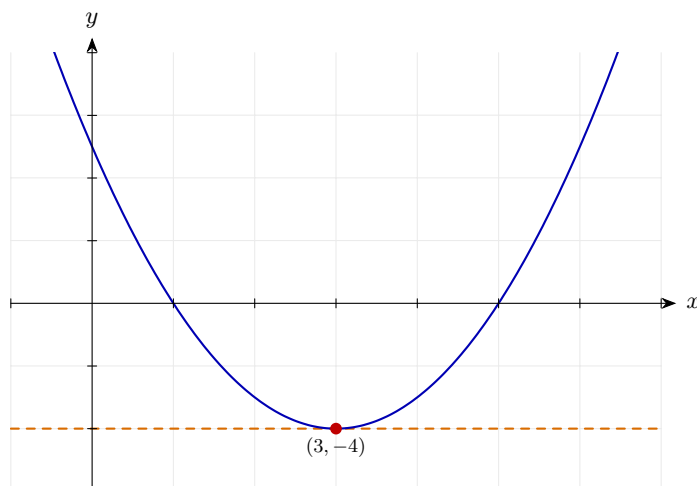
Step 3: Find the y -coordinate by substituting $x = 3$ into the original equation.

$$y = (3)^2 - 6(3) + 5 = 9 - 18 + 5 = -4$$

Always substitute back into y (not $\frac{dy}{dx}$) to find the y -coordinate.

Final answer: (3, -4)

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 19 (d2-019)

Difficulty: Foundation / Marks: 4

Find the coordinates of the points on $y = x^3 - 3x$ where the gradient is 0.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 3x^2 - 3$$

Apply the power rule term by term.

Step 2: Set equal to 0 and solve.

$$3x^2 - 3 = 0 \implies x^2 = 1 \implies x = \pm 1$$

Factorise or rearrange to find both values of x .

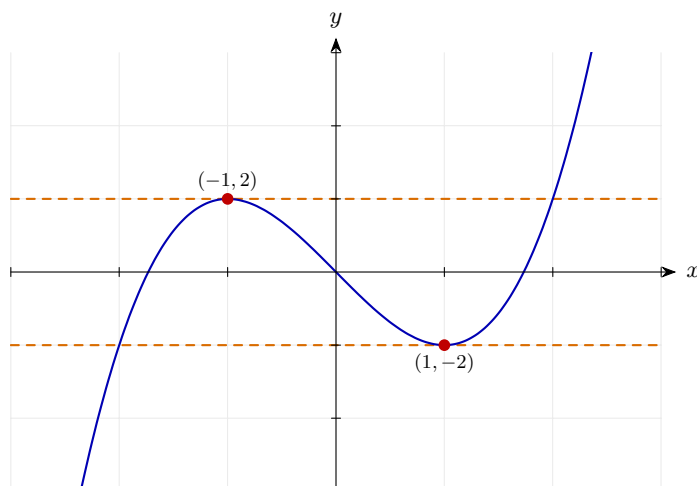
Step 3: Find the y -coordinates.

$$x = 1 : y = 1 - 3 = -2; \quad x = -1 : y = -1 + 3 = 2$$

Substitute each x back into $y = x^3 - 3x$.

Final answer: (1, -2) and (-1, 2)

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 20 (d2-020)

Difficulty: Foundation / Marks: 4

Find the equation of the tangent to $y = x^2 + 3x$ at the point $(1, 4)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Differentiate to find the gradient function.

$$\frac{dy}{dx} = 2x + 3$$

The derivative gives the gradient of the tangent at any point.

Step 2: Find the gradient at $x = 1$.

$$m = 2(1) + 3 = 5$$

Substitute $x = 1$ into the derivative.

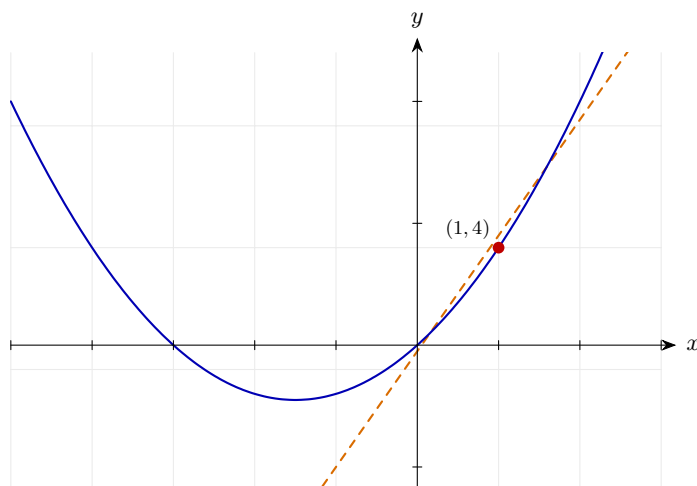
Step 3: Use $y - y_1 = m(x - x_1)$ with $m = 5$ and the point $(1, 4)$.

$$y - 4 = 5(x - 1) \implies y = 5x - 1$$

Rearrange to the required form $y = mx + c$.

Final answer: $y = 5x - 1$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 21 (d2-021)

Difficulty: Foundation / Marks: 4

Find the equation of the tangent to $y = 2x^3 - 5$ at the point $(1, -3)$. Give your answer in the form $ax + by + c = 0$, where a , b , and c are integers.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 6x^2$$

Apply the power rule to $2x^3$; the constant -5 differentiates to 0.

Step 2: Gradient at $x = 1$.

$$m = 6(1)^2 = 6$$

Substitute $x = 1$.

Step 3: Equation of tangent.

$$y - (-3) = 6(x - 1) \implies y = 6x - 9$$

Use point-slope form and expand.

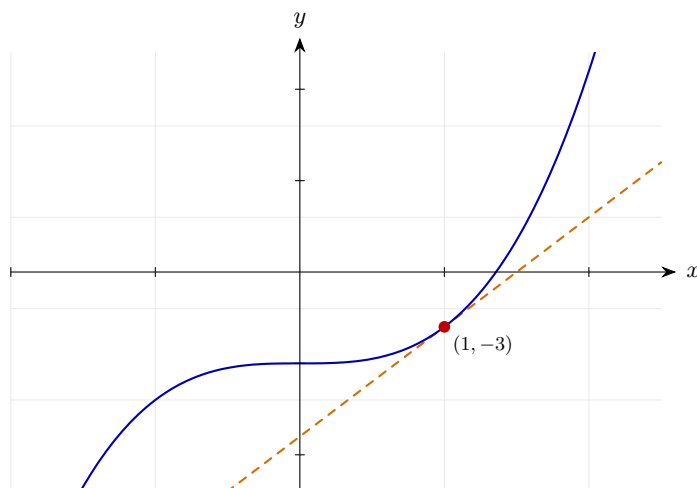
Step 4: Rearrange to the required form.

$$6x - y - 9 = 0$$

Move everything to one side so coefficients are integers.

Final answer: $6x - y - 9 = 0$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 22 (d2-022)

Difficulty: Foundation / Marks: 4

Find the equation of the normal to $y = x^2 - 4x + 3$ at the point $(3, 0)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Differentiate to find the gradient of the curve.

$$\frac{dy}{dx} = 2x - 4$$

The derivative gives the gradient of the tangent at any point.

Step 2: Find the gradient of the tangent at $x = 3$.

$$m_{\text{tangent}} = 2(3) - 4 = 2$$

Substitute $x = 3$ into the derivative.

Step 3: Find the gradient of the normal.

$$m_{\text{normal}} = -\frac{1}{2}$$

The normal is perpendicular to the tangent, so its gradient is the negative reciprocal: $-\frac{1}{m}$.

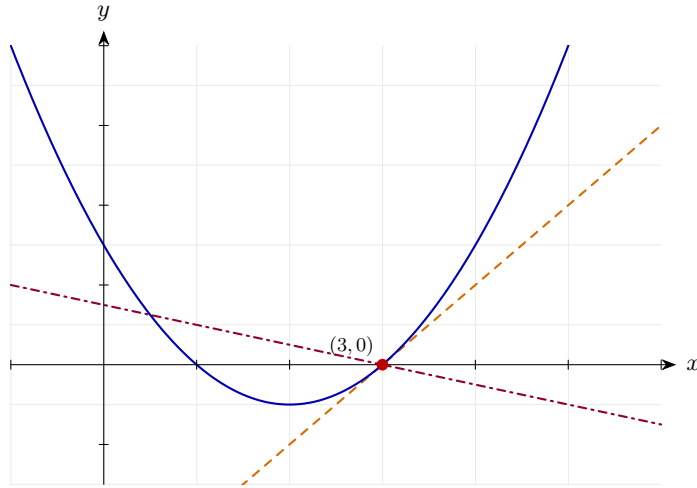
Step 4: Write the equation of the normal through $(3, 0)$.

$$y - 0 = -\frac{1}{2}(x - 3) \implies y = -\frac{1}{2}x + \frac{3}{2}$$

Use $y - y_1 = m(x - x_1)$ with the normal gradient and the given point.

Final answer: $y = -\frac{1}{2}x + \frac{3}{2}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 23 (d2-023)

Difficulty: Foundation / Marks: 4

Find the equation of the normal to $y = x^3 - 2x + 1$ at the point $(2, 5)$. Give your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 3x^2 - 2$$

Apply the power rule.

Step 2: Gradient of tangent at $x = 2$.

$$m_{\text{tangent}} = 3(2) - 2 = 10$$

$$3(2)^2 - 2 = 12 - 2 = 10.$$

Step 3: Gradient of normal.

$$m_{\text{normal}} = -\frac{1}{10}$$

Negative reciprocal of 10.

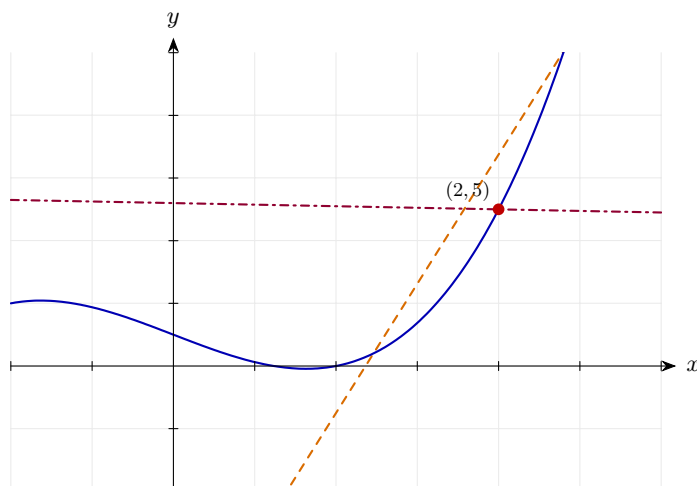
Step 4: Equation of normal through $(2, 5)$.

$$y - 5 = -\frac{1}{10}(x - 2) \implies 10y - 50 = -(x - 2) \implies x + 10y - 52 = 0$$

Multiply through by 10 to clear the fraction, then rearrange.

Final answer: $x + 10y - 52 = 0$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 24 (d2-024)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = (x + 2)(x - 5)$ and find the value of x where $f'(x) = 3$.

Worked solution.

Step 1: Expand the brackets.

$$f(x) = x^2 - 3x - 10$$

$$(x + 2)(x - 5) = x^2 - 5x + 2x - 10.$$

Step 2: Differentiate.

$$f'(x) = 2x - 3$$

Apply the power rule term by term.

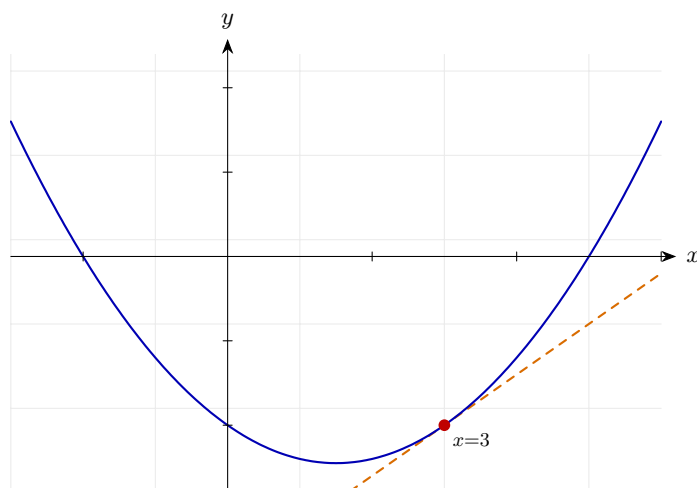
Step 3: Set $f'(x) = 3$ and solve.

$$2x - 3 = 3 \implies 2x = 6 \implies x = 3$$

Substitute the given gradient value and solve for x .

Final answer: $x = 3$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 25 (d2-025)

Difficulty: Foundation / Marks: 3

Differentiate $y = \frac{x^3 - 5x^2 + 2x}{x}$.

Worked solution.

Step 1: Divide each term in the numerator by x .

$$y = x^2 - 5x + 2$$

$$\frac{x^3}{x} = x^2, \frac{-5x^2}{x} = -5x, \frac{2x}{x} = 2.$$

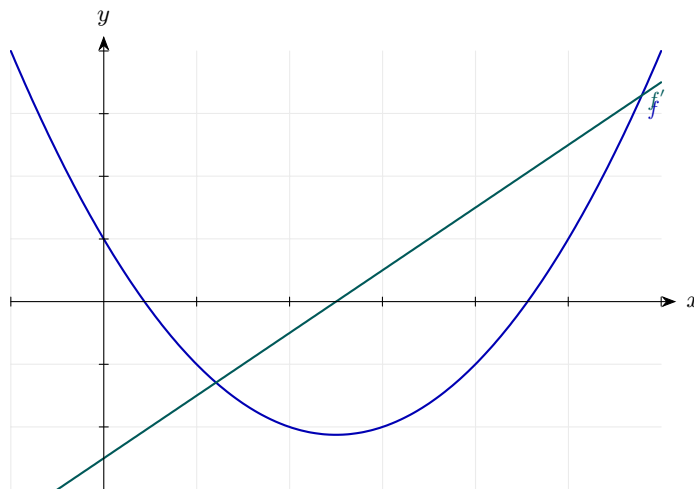
Step 2: Differentiate.

$$\frac{dy}{dx} = 2x - 5$$

The constant 2 differentiates to 0.

Final answer: $\frac{dy}{dx} = 2x - 5$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 26 (d2-026)

Difficulty: Foundation / Marks: 4

Find $\frac{dy}{dx}$ **when** $y = \frac{3x^3 + x}{\sqrt{x}}$.

Worked solution.

Step 1: Write $\sqrt{x} = x^{1/2}$ and divide each numerator term.

$$y = \frac{3x^3}{x^{1/2}} + \frac{x}{x^{1/2}} = 3x^{5/2} + x^{1/2}$$

Use the index law $\frac{x^a}{x^b} = x^{a-b}$.

Step 2: Differentiate.

$$\frac{dy}{dx} = 3 \cdot \frac{5}{2}x^{3/2} + \frac{1}{2}x^{-1/2} = \frac{15}{2}x^{3/2} + \frac{1}{2}x^{-1/2}$$

Apply the power rule to each term.

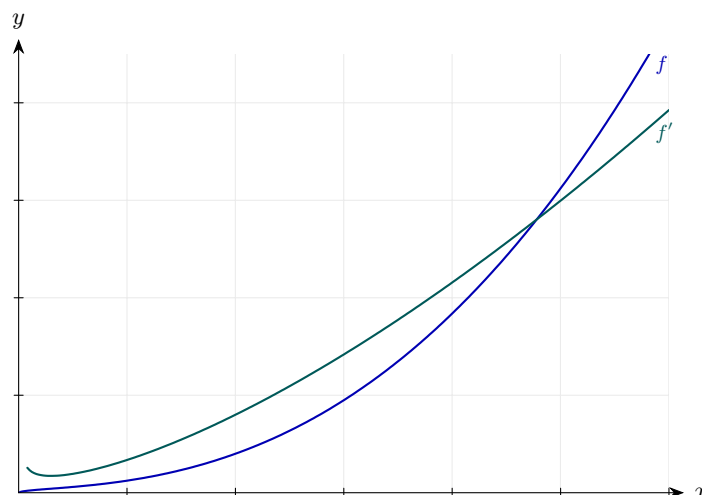
Step 3: Rewrite in a clean form.

$$\frac{dy}{dx} = \frac{15}{2}x\sqrt{x} + \frac{1}{2\sqrt{x}}$$

$$x^{3/2} = x \cdot \sqrt{x} \text{ and } x^{-1/2} = \frac{1}{\sqrt{x}}.$$

Final answer: $\frac{dy}{dx} = \frac{15x\sqrt{x}}{2} + \frac{1}{2\sqrt{x}}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 27 (d2-027)

Difficulty: Foundation / Marks: 3

Expand then differentiate $y = (3x - 1)^2$.

Worked solution.

Step 1: Expand the square.

$$(3x - 1)^2 = 9x^2 - 6x + 1$$

$$(a - b)^2 = a^2 - 2ab + b^2 \text{ with } a = 3x, b = 1.$$

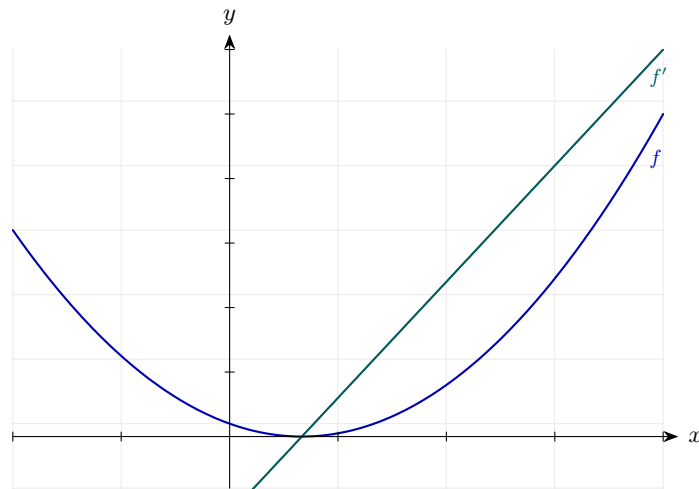
Step 2: Differentiate.

$$\frac{dy}{dx} = 18x - 6$$

Apply the power rule to each term; the constant 1 differentiates to 0.

Final answer: $\frac{dy}{dx} = 18x - 6$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 28 (d2-028)

Difficulty: Foundation / Marks: 4

Find the gradient of $y = \frac{x^2 - 4}{x}$ at the point $(-2, 0)$.

Worked solution.

Step 1: Simplify by splitting the fraction.

$$y = x - 4x^{-1}$$

$$\frac{x^2}{x} = x \text{ and } \frac{-4}{x} = -4x^{-1}.$$

Step 2: Differentiate.

$$\frac{dy}{dx} = 1 + 4x^{-2} = 1 + \frac{4}{x^2}$$

Derivative of $-4x^{-1}$ is $+4x^{-2}$ (multiply by -1 , reduce index by 1).

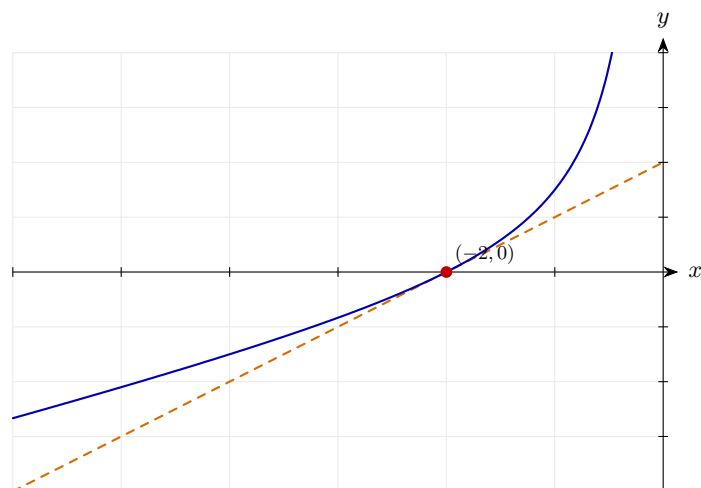
Step 3: Substitute $x = -2$.

$$1 + \frac{4}{(-2)^2} = 1 + \frac{4}{4} = 1 + 1 = 2$$

$$(-2)^2 = 4.$$

Final answer: Gradient = 2

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 29 (d2-029)

Difficulty: Foundation / Marks: 3

Differentiate $y = \sqrt{x}(x - 4)$.

Worked solution.

Step 1: Rewrite and expand.

$$y = x^{1/2}(x - 4) = x^{3/2} - 4x^{1/2}$$

Multiply $x^{1/2}$ by each term: $x^{1/2} \cdot x = x^{3/2}$ and $x^{1/2} \cdot (-4) = -4x^{1/2}$.

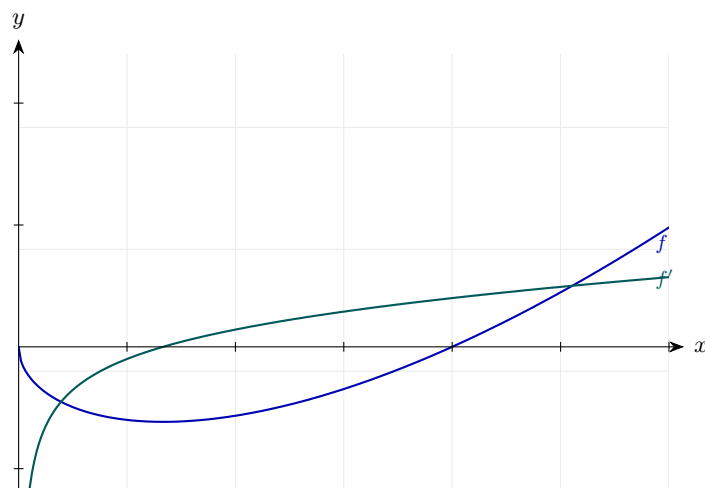
Step 2: Differentiate.

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} - 4 \cdot \frac{1}{2}x^{-1/2} = \frac{3}{2}\sqrt{x} - \frac{2}{\sqrt{x}}$$

Apply the power rule to each fractional-index term.

Final answer: $\frac{dy}{dx} = \frac{3\sqrt{x}}{2} - \frac{2}{\sqrt{x}}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 30 (d2-030)

Difficulty: Foundation / Marks: 4

Find the equation of the tangent to $y = (x-1)(x+3)$ at the point $(2, 5)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Expand the brackets.

$$y = x^2 + 2x - 3$$

$$(x-1)(x+3) = x^2 + 3x - x - 3.$$

Step 2: Differentiate.

$$\frac{dy}{dx} = 2x + 2$$

Power rule applied term by term.

Step 3: Gradient at $x = 2$.

$$m = 2(2) + 2 = 6$$

Substitute $x = 2$ into the derivative.

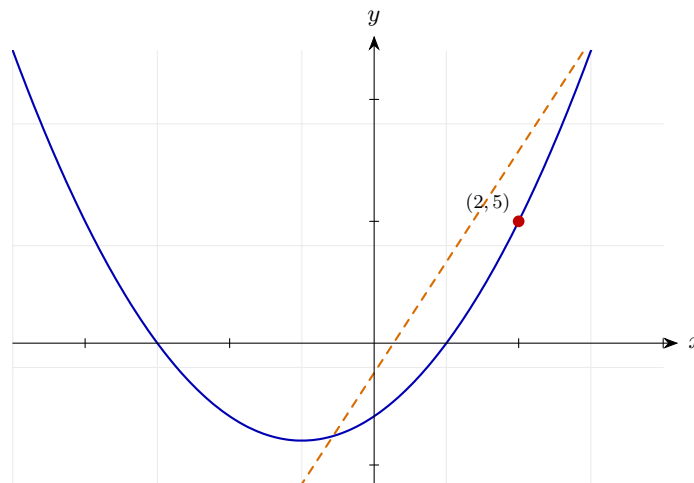
Step 4: Equation of tangent through $(2, 5)$.

$$y - 5 = 6(x - 2) \implies y = 6x - 7$$

Expand and rearrange to $y = mx + c$.

Final answer: $y = 6x - 7$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 31 (d2-031)

Difficulty: Foundation / Marks: 5

Find the equation of the normal to $y = 4\sqrt{x}$ at the point $(4, 8)$. Give your answer in the form $ax + by + c = 0$, where a , b , and c are integers.

Worked solution.

Step 1: Write as a power of x and differentiate.

$$y = 4x^{1/2} \implies \frac{dy}{dx} = 2x^{-1/2} = \frac{2}{\sqrt{x}}$$

Apply the power rule to $4x^{1/2}$.

Step 2: Gradient of tangent at $x = 4$.

$$m_{\text{tangent}} = \frac{2}{\sqrt{4}} = \frac{2}{2} = 1$$

Substitute $x = 4$.

Step 3: Gradient of normal.

$$m_{\text{normal}} = -1$$

Negative reciprocal of 1 is -1 .

Step 4: Equation of normal through $(4, 8)$.

$$y - 8 = -1(x - 4) \implies y = -x + 12$$

Point-slope form with $m = -1$.

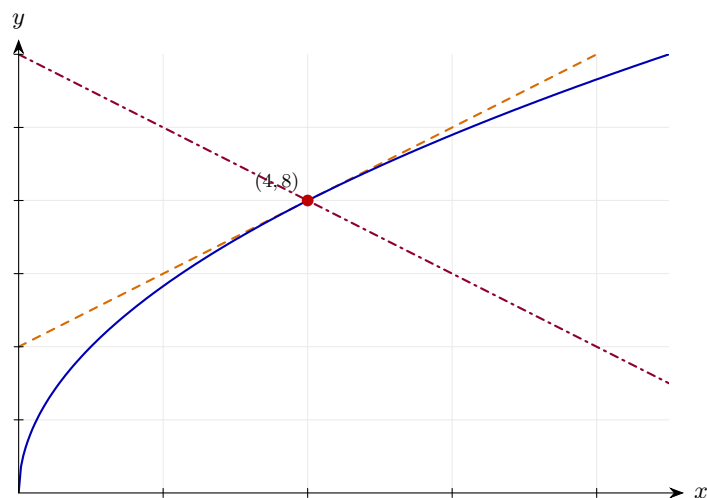
Step 5: Rearrange to integer form.

$$x + y - 12 = 0$$

Add x and subtract 12 from both sides.

Final answer: $x + y - 12 = 0$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 32 (d2-032)

Difficulty: Foundation / Marks: 6

The curve C has equation $y = x^3 - 4x + 3$. (a) Find $\frac{dy}{dx}$. (b) Show that the point $P(2, 3)$ lies on C . (c) Find the equation of the tangent to C at P , giving your answer in the form $y = mx + c$.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{dy}{dx} = 3x^2 - 4$$

Apply the power rule to each term.

Step 2: (b) Substitute $x = 2$ into $y = x^3 - 4x + 3$.

$$y = (2)^3 - 4(2) + 3 = 8 - 8 + 3 = 3 \checkmark$$

The point $(2, 3)$ satisfies the equation, so it lies on C .

Step 3: (c) Gradient at $P(2, 3)$.

$$m = 3(2)^2 - 4 = 12 - 4 = 8$$

Substitute $x = 2$ into the derivative.

Step 4: Write the tangent equation.

$$y - 3 = 8(x - 2) \implies y = 8x - 13$$

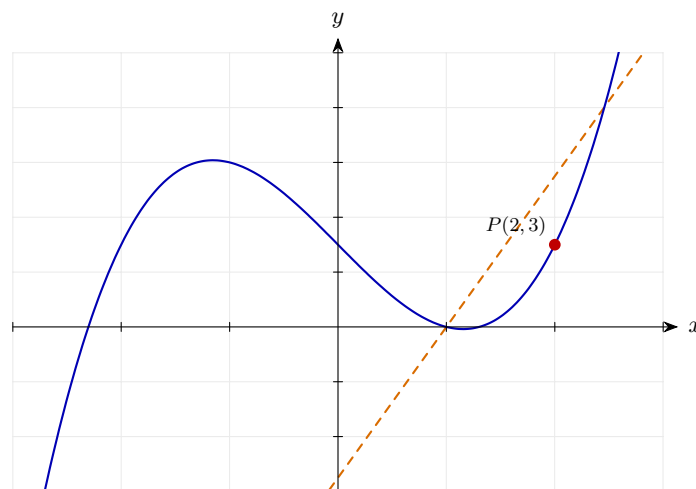
Use point-slope form and rearrange.

Final answer: (a) $\frac{dy}{dx} = 3x^2 - 4$

(b) $y = 8 - 8 + 3 = 3 \checkmark$

(c) $y = 8x - 13$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 33 (d2-033)

Difficulty: Foundation / Marks: 6

The curve C has equation $y = 2x^2 - 7x + 4$. (a) Find the gradient of C at the point $A(3, 1)$.
(b) Hence find the equation of the normal to C at A , giving your answer in the form $ax + by + c = 0$.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{dy}{dx} = 4x - 7$$

Power rule applied to each term.

Step 2: Gradient at $x = 3$.

$$m_{\text{tangent}} = 4(3) - 7 = 5$$

Substitute $x = 3$.

Step 3: (b) Gradient of normal.

$$m_{\text{normal}} = -\frac{1}{5}$$

The normal gradient is the negative reciprocal.

Step 4: Equation of normal through $(3, 1)$.

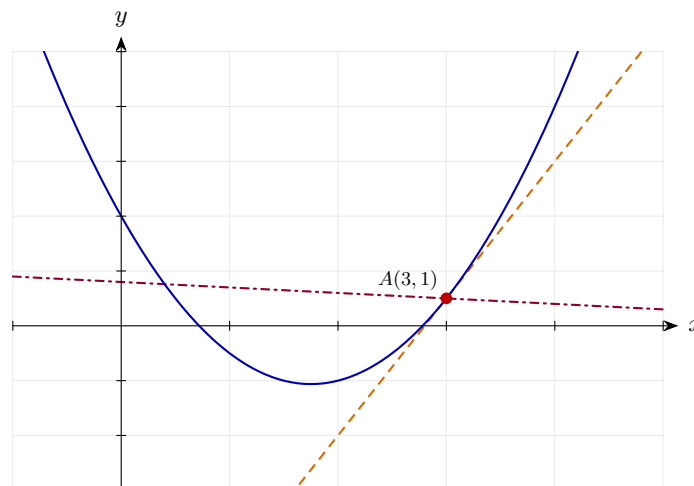
$$y - 1 = -\frac{1}{5}(x - 3) \implies 5y - 5 = -(x - 3) \implies x + 5y - 8 = 0$$

Multiply through by 5 to clear the fraction, then rearrange.

Final answer: (a) Gradient = 5

(b) $x + 5y - 8 = 0$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 34 (d2-034)

Difficulty: Foundation / Marks: 8

The function $f(x) = x^3 - 6x^2 + 9x + 2$. (a) Find $f'(x)$. (b) Find the coordinates of the two stationary points of the curve $y = f(x)$. (c) Find the equation of the tangent to the curve at the point where $x = 4$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: (a) Differentiate $f(x)$.

$$f'(x) = 3x^2 - 12x + 9$$

Apply the power rule to each term.

Step 2: (b) Set $f'(x) = 0$.

$$3x^2 - 12x + 9 = 0 \implies x^2 - 4x + 3 = 0 \implies (x - 1)(x - 3) = 0$$

Divide through by 3, then factorise.

Step 3: So $x = 1$ or $x = 3$. Find the y -coordinates.

$$f(1) = 1 - 6 + 9 + 2 = 6; \quad f(3) = 27 - 54 + 27 + 2 = 2$$

Substitute each x back into $f(x)$.

Step 4: (c) Gradient at $x = 4$.

$$f'(4) = 3(16) - 12(4) + 9 = 48 - 48 + 9 = 9$$

Substitute $x = 4$ into $f'(x)$.

Step 5: Point on curve when $x = 4$.

$$f(4) = 64 - 96 + 36 + 2 = 6$$

Substitute into $f(x)$ to find the y -coordinate.

Step 6: Tangent at $(4, 6)$ with gradient 9.

$$y - 6 = 9(x - 4) \implies y = 9x - 30$$

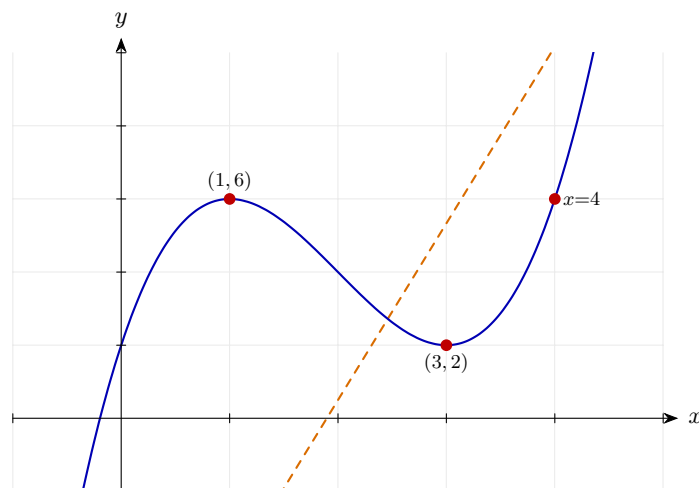
Use point-slope form and rearrange.

Final answer: (a) $f'(x) = 3x^2 - 12x + 9$

(b) $(1, 6)$ and $(3, 2)$

(c) $y = 9x - 30$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 35 (d2-035)

Difficulty: Foundation / Marks: 8

The curve C has equation $y = f(x)$, where $f(x) = \frac{x^3 - 4x}{\sqrt{x}}$ for $x > 0$. (a) Show that $f(x)$ can be written in the form $ax^p + bx^q$, stating the values of a , p , b , and q . (b) Find $f'(x)$. (c) Find the equation of the tangent to C at the point where $x = 4$, giving your answer in the form $y = mx + c$.

Worked solution.

Step 1: (a) Write $\sqrt{x} = x^{1/2}$ and divide each term.

$$f(x) = \frac{x^3}{x^{1/2}} - \frac{4x}{x^{1/2}} = x^{5/2} - 4x^{1/2}$$

Use $\frac{x^a}{x^b} = x^{a-b}$: $x^{3-1/2} = x^{5/2}$ and $x^{1-1/2} = x^{1/2}$.

Step 2: State the constants.

$$a = 1, p = \frac{5}{2}, b = -4, q = \frac{1}{2}$$

Match the form $ax^p + bx^q$.

Step 3: (b) Differentiate $f(x) = x^{5/2} - 4x^{1/2}$.

$$f'(x) = \frac{5}{2}x^{3/2} - 4 \cdot \frac{1}{2}x^{-1/2} = \frac{5}{2}x^{3/2} - \frac{2}{\sqrt{x}}$$

Apply the power rule to each term.

Step 4: (c) Find the y -coordinate when $x = 4$.

$$f(4) = 4^{5/2} - 4 \cdot 4^{1/2} = 32 - 8 = 24$$

$$4^{5/2} = (\sqrt{4})^5 = 2^5 = 32 \text{ and } 4^{1/2} = 2.$$

Step 5: Gradient at $x = 4$.

$$f'(4) = \frac{5}{2}(4)^{3/2} - \frac{2}{\sqrt{4}} = \frac{5}{2}(8) - 1 = 20 - 1 = 19$$

$$4^{3/2} = (\sqrt{4})^3 = 8.$$

Step 6: Tangent through $(4, 24)$ with gradient 19.

$$y - 24 = 19(x - 4) \implies y = 19x - 52$$

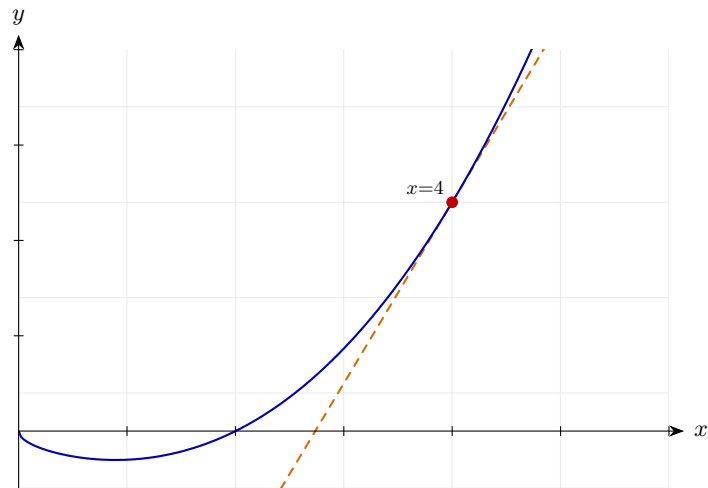
Point-slope form, expanded and rearranged.

Final answer: (a) $f(x) = x^{5/2} - 4x^{1/2}$; $a = 1, p = \frac{5}{2}, b = -4, q = \frac{1}{2}$

(b) $f'(x) = \frac{5}{2}x^{3/2} - \frac{2}{\sqrt{x}}$

(c) $y = 19x - 52$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 36 (d2-036)

Difficulty: Standard / Marks: 2

Differentiate $y = \frac{5}{x^2}$.

Worked solution.

Step 1: Rewrite with a negative index.

$$y = 5x^{-2}$$

Bring the power to the numerator by negating the index.

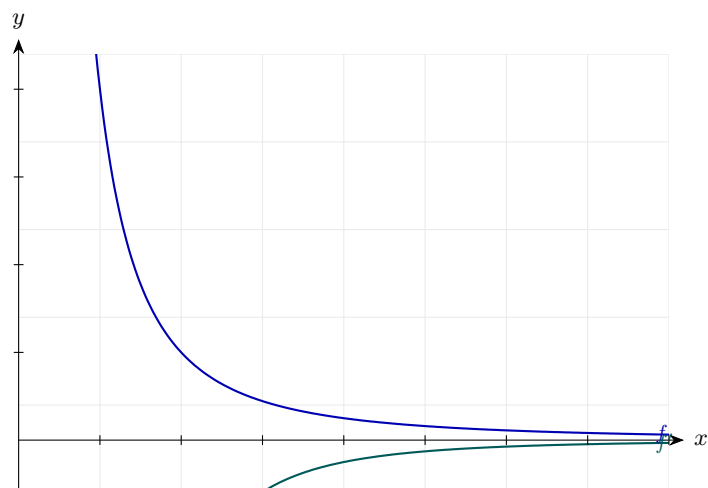
Step 2: Apply the power rule.

$$\frac{dy}{dx} = -10x^{-3} = -\frac{10}{x^3}$$

Multiply by the index (-2) and reduce by 1.

Final answer: $\frac{dy}{dx} = -\frac{10}{x^3}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 37 (d2-037)

Difficulty: Standard / Marks: 2

Differentiate $f(x) = \frac{2}{\sqrt{x}}$.

Worked solution.

Step 1: Rewrite as a power.

$$f(x) = 2x^{-1/2}$$

$$\frac{1}{\sqrt{x}} = x^{-1/2}.$$

Step 2: Differentiate.

$$f'(x) = 2 \cdot \left(-\frac{1}{2}\right) x^{-3/2} = -x^{-3/2}$$

Apply the power rule.

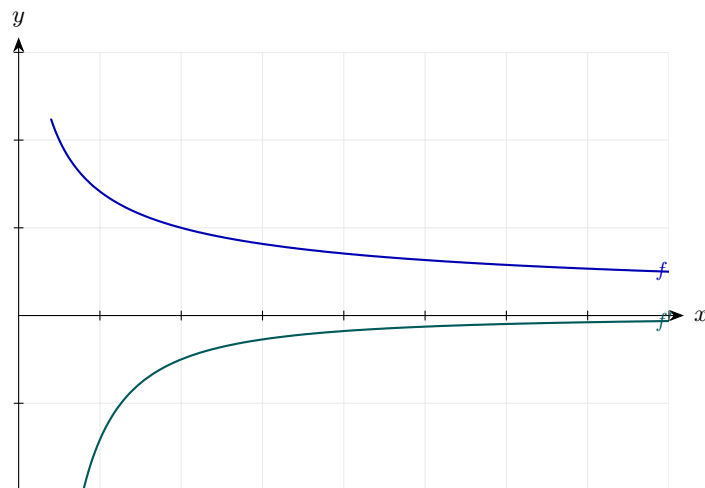
Step 3: Rewrite with a surd.

$$f'(x) = -\frac{1}{x\sqrt{x}}$$

$$x^{-3/2} = \frac{1}{x^{3/2}} = \frac{1}{x\sqrt{x}}.$$

Final answer: $f'(x) = -\frac{1}{x\sqrt{x}}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 38 (d2-038)

Difficulty: Standard / Marks: 2

Differentiate $y = 3\sqrt[3]{x}$.

Worked solution.

Step 1: Rewrite using a fractional index.

$$y = 3x^{1/3}$$

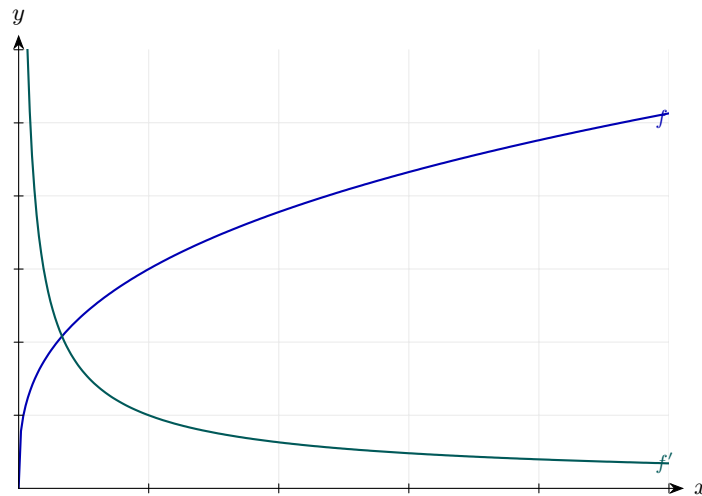
$$\sqrt[3]{x} = x^{1/3}.$$

Step 2: Apply the power rule.

$$\frac{dy}{dx} = 3 \cdot \frac{1}{3}x^{-2/3} = x^{-2/3} = \frac{1}{x^{2/3}}$$

The coefficients cancel neatly.

Final answer: $\frac{dy}{dx} = \frac{1}{\sqrt[3]{x^2}}$

Diagram.

Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 39 (d2-039)

Difficulty: Standard / Marks: 3

Differentiate $y = 6x^{2/3} - 4x^{-1}$.

Worked solution.

Step 1: Differentiate term by term.

$$\frac{dy}{dx} = 6 \cdot \frac{2}{3}x^{-1/3} - 4 \cdot (-1)x^{-2}$$

Apply the power rule to each term.

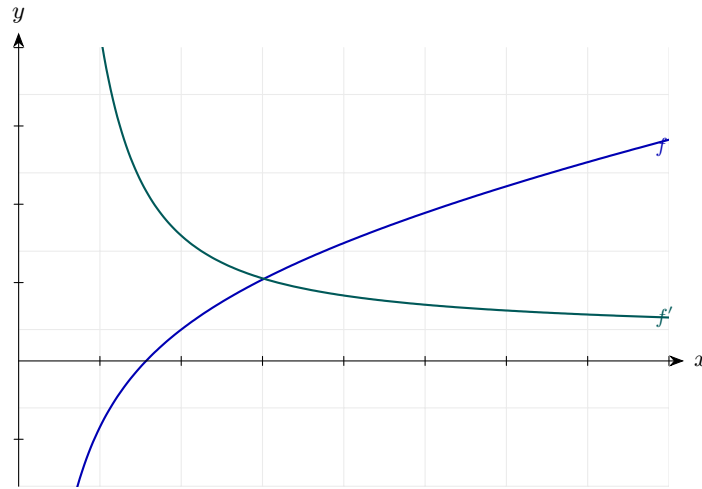
Step 2: Simplify.

$$= 4x^{-1/3} + 4x^{-2} = \frac{4}{\sqrt[3]{x}} + \frac{4}{x^2}$$

Rewrite with positive indices.

Final answer: $\frac{dy}{dx} = \frac{4}{\sqrt[3]{x}} + \frac{4}{x^2}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 40 (d2-040)

Difficulty: Standard / Marks: 3

Differentiate $f(x) = \frac{x^2 + 1}{x^3}$.

Worked solution.

Step 1: Split the fraction.

$$f(x) = \frac{x^2}{x^3} + \frac{1}{x^3} = x^{-1} + x^{-3}$$

Divide each numerator term by x^3 and write as powers.

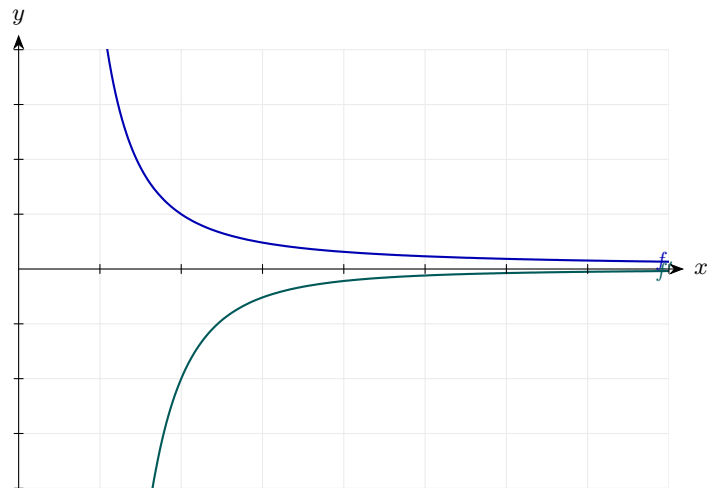
Step 2: Differentiate.

$$f'(x) = -x^{-2} - 3x^{-4} = -\frac{1}{x^2} - \frac{3}{x^4}$$

Apply the power rule to each term.

Final answer: $f'(x) = -\frac{1}{x^2} - \frac{3}{x^4}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 41 (d2-041)

Difficulty: Standard / Marks: 3

Differentiate $y = \frac{2x^{3/2} - 5}{\sqrt{x}}$.

Worked solution.

Step 1: Split the fraction and simplify each term.

$$y = \frac{2x^{3/2}}{x^{1/2}} - \frac{5}{x^{1/2}} = 2x - 5x^{-1/2}$$

Use the index law $\frac{x^a}{x^b} = x^{a-b}$.

Step 2: Differentiate.

$$\frac{dy}{dx} = 2 + \frac{5}{2}x^{-3/2}$$

Apply the power rule.

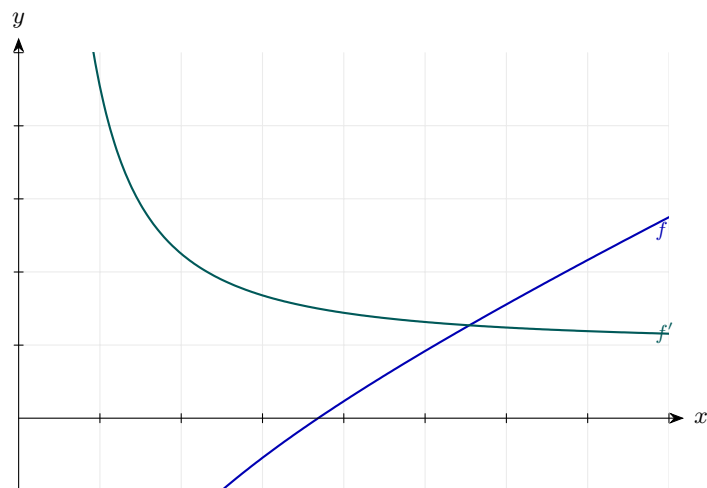
Step 3: Rewrite.

$$= 2 + \frac{5}{2x\sqrt{x}}$$

Convert the negative index back to a surd.

Final answer: $\frac{dy}{dx} = 2 + \frac{5}{2x\sqrt{x}}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 42 (d2-042)

Difficulty: Standard / Marks: 3

Differentiate $y = (x^2 - 3)(x + 1)$.

Worked solution.

Step 1: Expand the brackets.

$$y = x^3 + x^2 - 3x - 3$$

Multiply term by term.

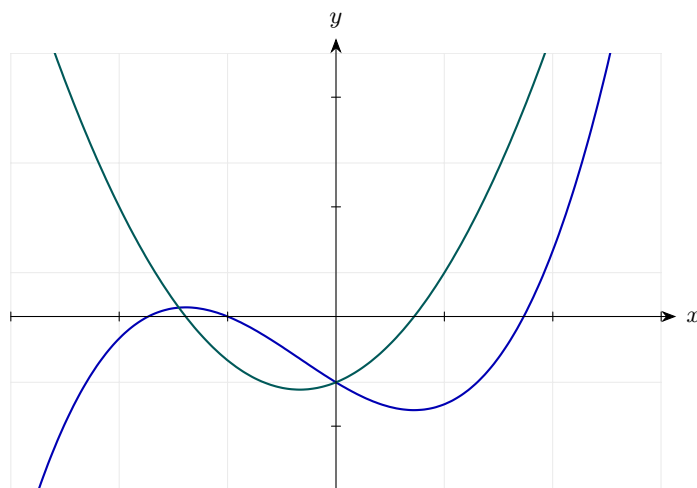
Step 2: Differentiate.

$$\frac{dy}{dx} = 3x^2 + 2x - 3$$

Differentiate each term; constants vanish.

Final answer: $\frac{dy}{dx} = 3x^2 + 2x - 3$

Diagram.

f' f 

Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 43 (d2-043)

Difficulty: Standard / Marks: 4

Expand and differentiate $y = (2x - 1)^2(x + 3)$.

Worked solution.

Step 1: Expand $(2x - 1)^2$ first.

$$(2x - 1)^2 = 4x^2 - 4x + 1$$

Standard square expansion.

Step 2: Multiply by $(x + 3)$.

$$(4x^2 - 4x + 1)(x + 3) = 4x^3 + 12x^2 - 4x^2 - 12x + x + 3 = 4x^3 + 8x^2 - 11x + 3$$

Expand term by term then combine like terms.

Step 3: Differentiate.

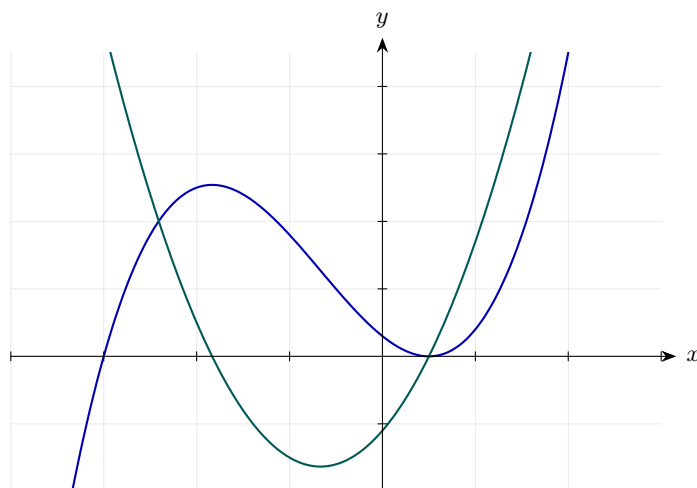
$$\frac{dy}{dx} = 12x^2 + 16x - 11$$

Apply the power rule.

Final answer: $\frac{dy}{dx} = 12x^2 + 16x - 11$

Diagram.

f'



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 44 (d2-044)

Difficulty: Standard / Marks: 3

Differentiate $y = \sqrt{x}(x^2 - 4)$.

Worked solution.

Step 1: Expand using $\sqrt{x} = x^{1/2}$.

$$y = x^{1/2} \cdot x^2 - 4x^{1/2} = x^{5/2} - 4x^{1/2}$$

Use the index law $x^a \cdot x^b = x^{a+b}$.

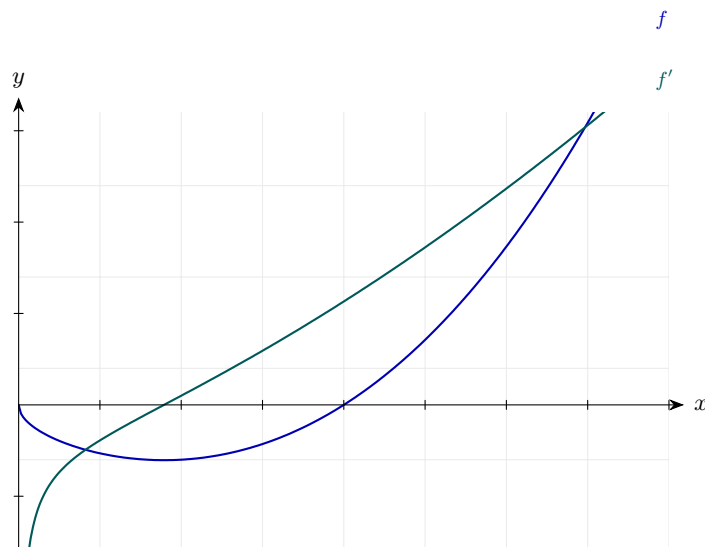
Step 2: Differentiate term by term.

$$\frac{dy}{dx} = \frac{5}{2}x^{3/2} - 2x^{-1/2} = \frac{5}{2}x\sqrt{x} - \frac{2}{\sqrt{x}}$$

Apply the power rule and convert back to surd form.

Final answer: $\frac{dy}{dx} = \frac{5}{2}x\sqrt{x} - \frac{2}{\sqrt{x}}$
--

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 45 (d2-045)

Difficulty: Standard / Marks: 4

Differentiate $f(x) = \frac{(x+2)(x-1)}{x^2}$.

Worked solution.

Step 1: Expand the numerator.

$$(x+2)(x-1) = x^2 + x - 2$$

Standard expansion.

Step 2: Divide each term by x^2 .

$$f(x) = 1 + x^{-1} - 2x^{-2}$$

Write each quotient as a power of x .

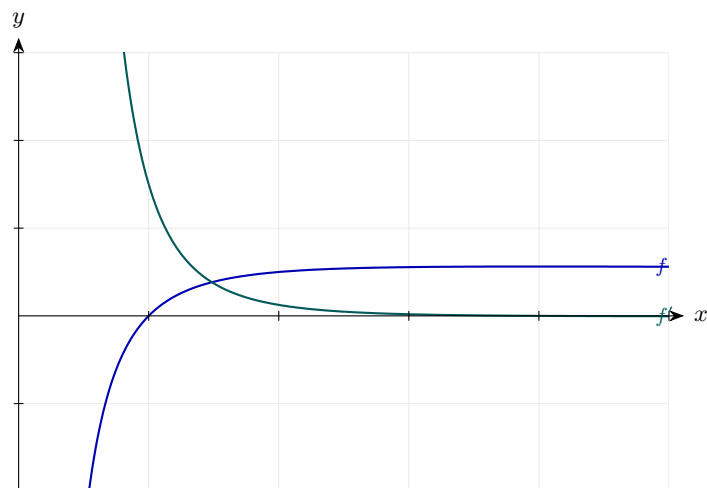
Step 3: Differentiate.

$$f'(x) = -x^{-2} + 4x^{-3} = -\frac{1}{x^2} + \frac{4}{x^3}$$

Constants vanish; apply the power rule to the remaining terms.

Final answer: $f'(x) = -\frac{1}{x^2} + \frac{4}{x^3}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 46 (d2-046)

Difficulty: Standard / Marks: 4

Differentiate $y = \left(x + \frac{1}{x}\right)^2$.

Worked solution.

Step 1: Expand the square.

$$y = x^2 + 2 + \frac{1}{x^2} = x^2 + 2 + x^{-2}$$

Use $(a + b)^2 = a^2 + 2ab + b^2$; note that $x \cdot \frac{1}{x} = 1$.

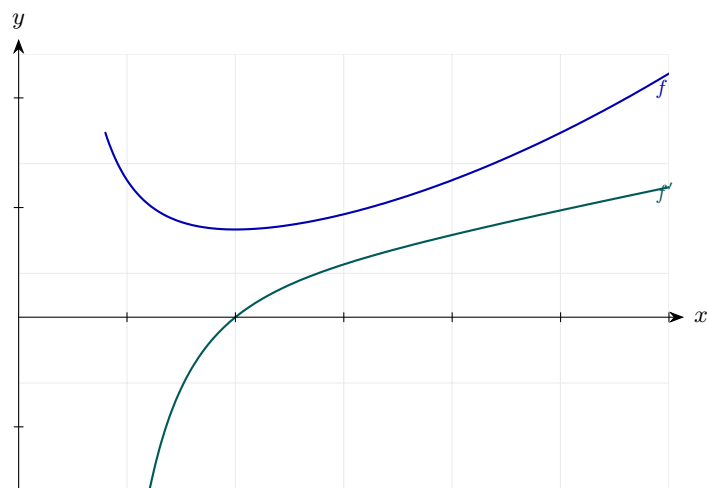
Step 2: Differentiate.

$$\frac{dy}{dx} = 2x - 2x^{-3} = 2x - \frac{2}{x^3}$$

Constant disappears; apply the power rule.

Final answer: $\frac{dy}{dx} = 2x - \frac{2}{x^3}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 47 (d2-047)

Difficulty: Standard / Marks: 4

Differentiate $y = \frac{(\sqrt{x} + 1)^2}{x}$.

Worked solution.

Step 1: Expand the numerator.

$$(\sqrt{x} + 1)^2 = x + 2\sqrt{x} + 1$$

$$(\sqrt{x})^2 = x.$$

Step 2: Divide each term by x .

$$y = 1 + 2x^{-1/2} + x^{-1}$$

$$\frac{\sqrt{x}}{x} = x^{-1/2}.$$

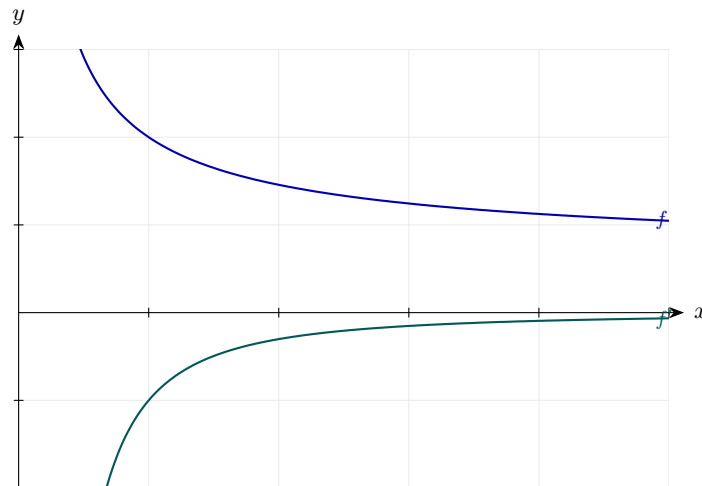
Step 3: Differentiate.

$$\frac{dy}{dx} = -x^{-3/2} - x^{-2} = -\frac{1}{x\sqrt{x}} - \frac{1}{x^2}$$

Apply the power rule.

Final answer: $\frac{dy}{dx} = -\frac{1}{x\sqrt{x}} - \frac{1}{x^2}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 48 (d2-048)

Difficulty: Standard / Marks: 3

Differentiate $f(x) = (x^2 + 2x)(x - 3)$.

Worked solution.*Step 1: Expand the brackets.*

$$f(x) = x^3 - 3x^2 + 2x^2 - 6x = x^3 - x^2 - 6x$$

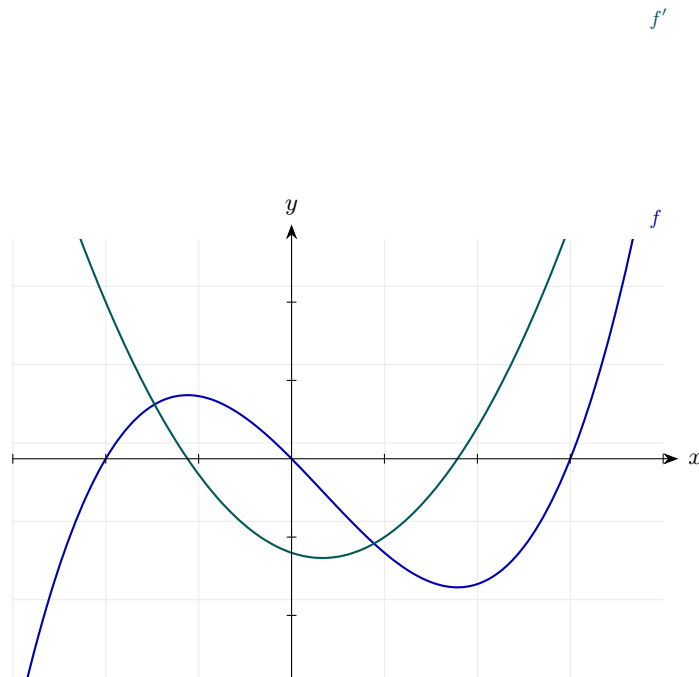
Collect like terms.

Step 2: Differentiate.

$$f'(x) = 3x^2 - 2x - 6$$

Apply the power rule.

Final answer: $f'(x) = 3x^2 - 2x - 6$

Diagram.

Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 49 (d2-049)*Difficulty: Standard / Marks: 3***Find the gradient of the curve $y = x^3 - 2x^2 + 5$ at the point where $x = -1$.****Worked solution.***Step 1: Differentiate.*

$$\frac{dy}{dx} = 3x^2 - 4x$$

Apply the power rule.

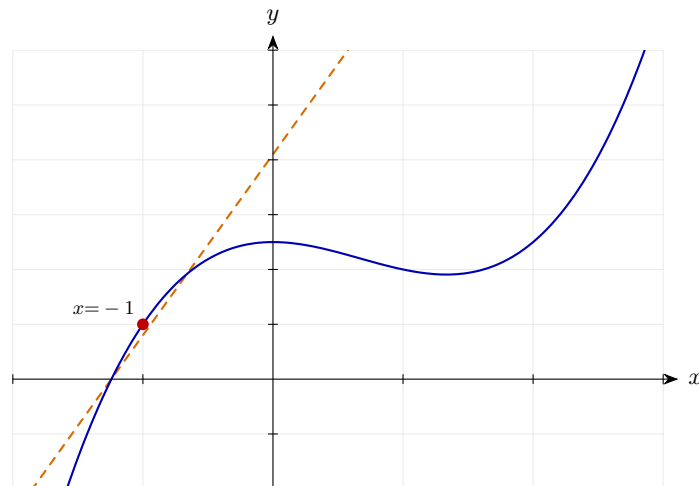
Step 2: Substitute $x = -1$.

$$3(-1)^2 - 4(-1) = 3 + 4 = 7$$

Evaluate the gradient function.

Final answer: Gradient at $x = -1$ is 7.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 50 (d2-050)

Difficulty: Standard / Marks: 3

Find the gradient of $y = \frac{4}{x} + x^2$ at the point where $x = 2$.

Worked solution.

Step 1: Rewrite and differentiate.

$$y = 4x^{-1} + x^2 \Rightarrow \frac{dy}{dx} = -4x^{-2} + 2x$$

Apply the power rule to each term.

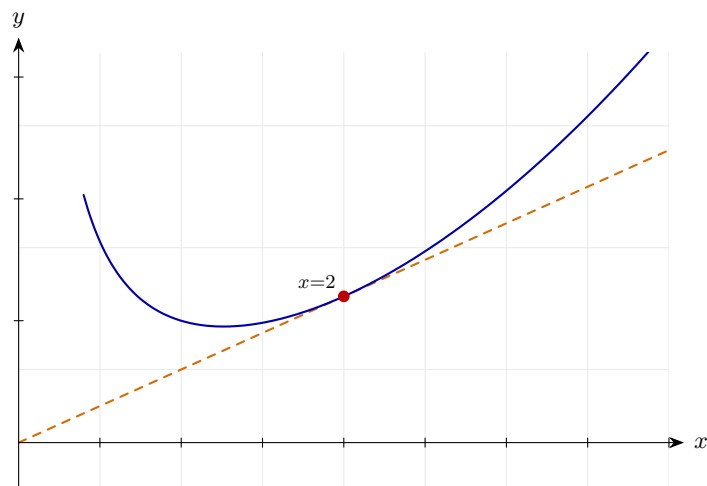
Step 2: Substitute $x = 2$.

$$-\frac{4}{4} + 4 = -1 + 4 = 3$$

Evaluate.

Final answer: Gradient at $x = 2$ is 3.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 51 (d2-051)

Difficulty: Standard / Marks: 4

Find the gradient of $f(x) = \sqrt{x}(x + 2)$ at the point where $x = 4$.

Worked solution.

Step 1: Expand.

$$f(x) = x^{3/2} + 2x^{1/2}$$

Write as powers of x .

Step 2: Differentiate.

$$f'(x) = \frac{3}{2}x^{1/2} + x^{-1/2} = \frac{3}{2}\sqrt{x} + \frac{1}{\sqrt{x}}$$

Apply the power rule.

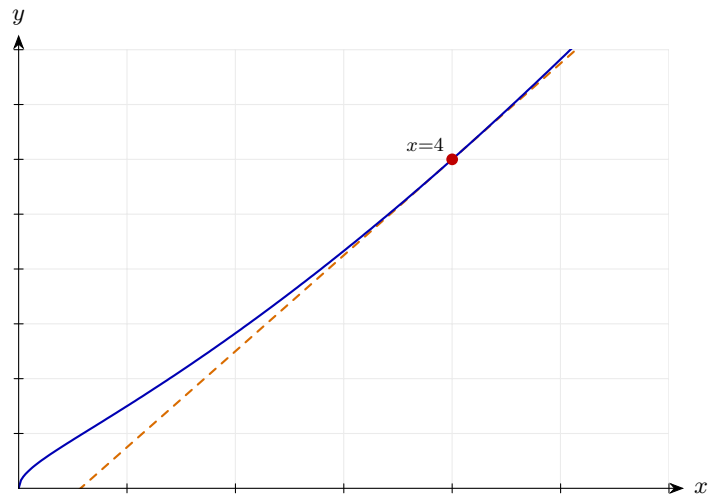
Step 3: Substitute $x = 4$.

$$f'(4) = \frac{3}{2}(2) + \frac{1}{2} = 3 + \frac{1}{2} = \frac{7}{2}$$

$$\sqrt{4} = 2, \frac{1}{\sqrt{4}} = \frac{1}{2}.$$

Final answer: Gradient at $x = 4$ is $\frac{7}{2}$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 52 (d2-052)

Difficulty: Standard / Marks: 4

The curve C has equation $y = x^3 - 4x$. Find the values of x at which the gradient of C is 8.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 3x^2 - 4$$

Apply the power rule.

Step 2: Set gradient equal to 8.

$$3x^2 - 4 = 8 \Rightarrow 3x^2 = 12 \Rightarrow x^2 = 4$$

Rearrange and solve.

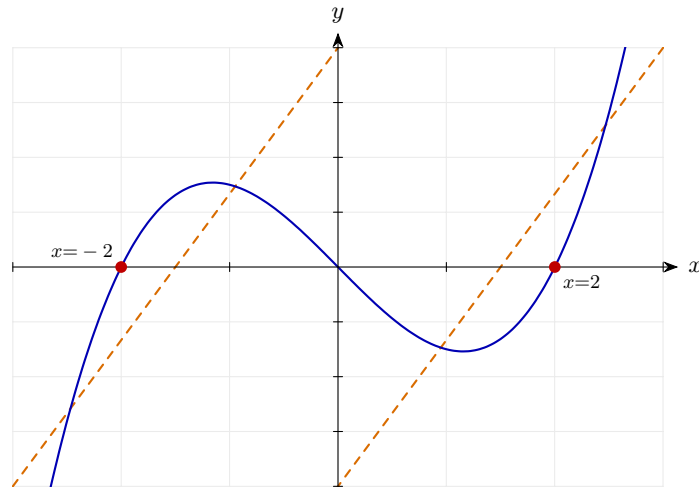
Step 3: Solve.

$$x = \pm 2$$

Two values of x .

Final answer: $x = 2$ or $x = -2$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 53 (d2-053)

Difficulty: Standard / Marks: 4

Find the coordinates of the point on $y = 2x^2 - 3x + 1$ where the gradient is 5.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 4x - 3$$

Apply the power rule.

Step 2: Set the derivative equal to 5.

$$4x - 3 = 5 \Rightarrow x = 2$$

Solve for x .

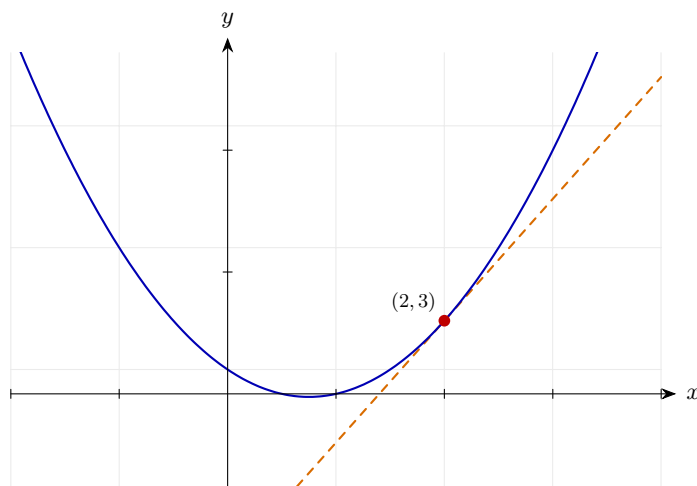
Step 3: Find y .

$$y = 2(4) - 3(2) + 1 = 8 - 6 + 1 = 3$$

Substitute into the original equation.

Final answer: Point is $(2, 3)$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 54 (d2-054)

Difficulty: Standard / Marks: 4

Find the equation of the tangent to $y = x^3 - 2x$ at the point $(2, 4)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 3x^2 - 2$$

Apply the power rule.

Step 2: Evaluate at $x = 2$.

$$3(4) - 2 = 10$$

The tangent has gradient 10.

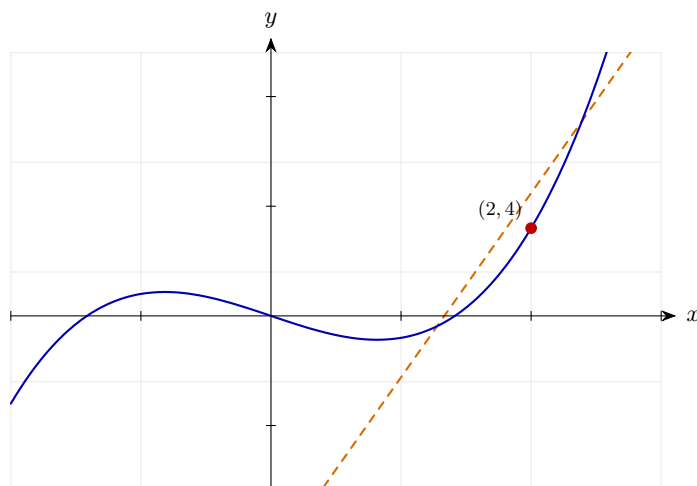
Step 3: Use point-slope form.

$$y - 4 = 10(x - 2) \Rightarrow y = 10x - 16$$

Expand and rearrange.

Final answer: $y = 10x - 16$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 55 (d2-055)

Difficulty: Standard / Marks: 4

Find the equation of the tangent to $y = \frac{6}{x}$ at the point where $x = 3$.

Worked solution.

Step 1: Find the y -coordinate.

$$y = \frac{6}{3} = 2 \Rightarrow (3, 2)$$

Substitute $x = 3$.

Step 2: Differentiate.

$$y = 6x^{-1} \Rightarrow \frac{dy}{dx} = -6x^{-2} = -\frac{6}{x^2}$$

Use the power rule.

Step 3: Gradient at $x = 3$.

$$-\frac{6}{9} = -\frac{2}{3}$$

Evaluate the derivative.

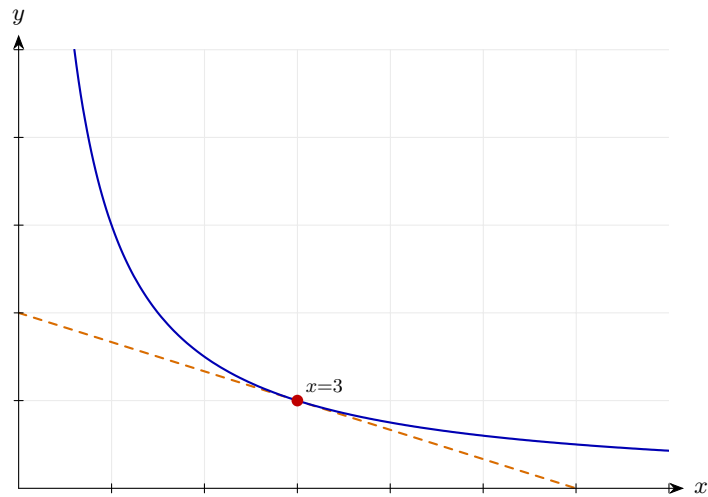
Step 4: Tangent equation.

$$y - 2 = -\frac{2}{3}(x - 3) \Rightarrow y = -\frac{2}{3}x + 4$$

Point-slope, then rearrange.

Final answer: $y = -\frac{2}{3}x + 4$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 56 (d2-056)

Difficulty: Standard / Marks: 4

Find the equation of the normal to $y = x^2 + 2x - 3$ at the point $(1, 0)$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 2x + 2$$

Apply the power rule.

Step 2: Gradient of tangent at $(1, 0)$.

$$2(1) + 2 = 4$$

Evaluate the derivative.

Step 3: Gradient of normal.

$$m_{\text{normal}} = -\frac{1}{4}$$

Perpendicular: $m_1 m_2 = -1$.

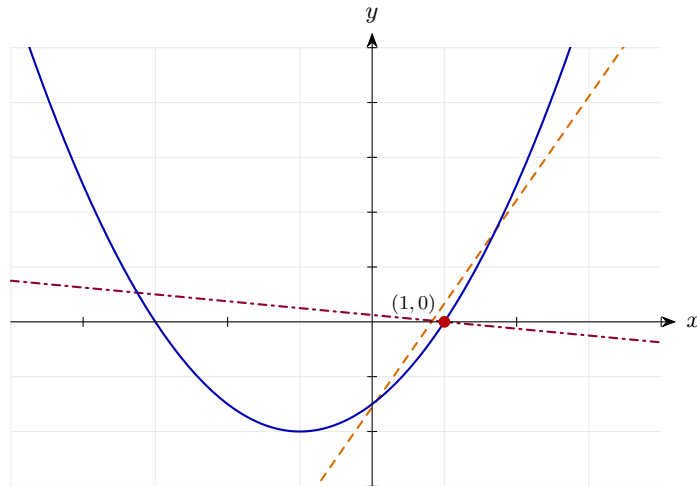
Step 4: Normal equation.

$$y - 0 = -\frac{1}{4}(x - 1) \Rightarrow y = -\frac{1}{4}x + \frac{1}{4}$$

Point-slope, then rearrange.

Final answer: $y = -\frac{1}{4}x + \frac{1}{4}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 57 (d2-057)

Difficulty: Standard / Marks: 4

The curve C has equation $y = \sqrt{x}$. Find the equation of the tangent to C at the point where $x = 9$. Give your answer in the form $y = mx + c$.

Worked solution.

Step 1: Find the y-coordinate.

$$y = \sqrt{9} = 3 \Rightarrow (9, 3)$$

Substitute $x = 9$.

Step 2: Differentiate.

$$y = x^{1/2} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

Apply the power rule.

Step 3: Gradient at $x = 9$.

$$\frac{1}{2 \cdot 3} = \frac{1}{6}$$

Evaluate.

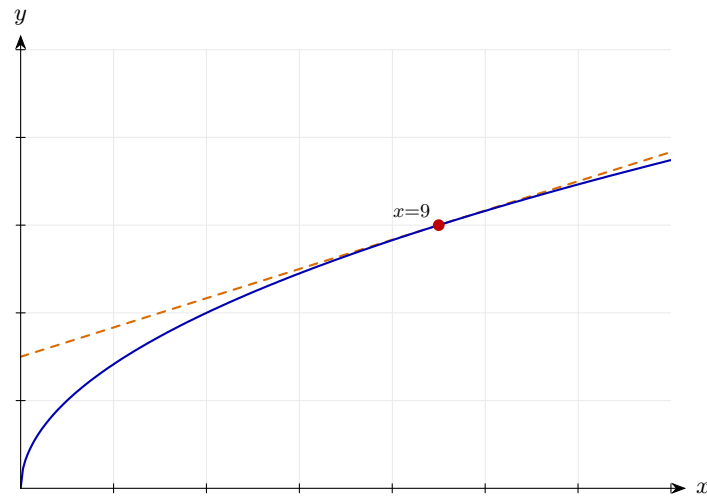
Step 4: Tangent equation.

$$y - 3 = \frac{1}{6}(x - 9) \Rightarrow y = \frac{1}{6}x + \frac{3}{2}$$

Point-slope, then rearrange.

Final answer: $y = \frac{1}{6}x + \frac{3}{2}$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 58 (d2-058)

Difficulty: Standard / Marks: 5

Find the equation of the normal to $y = 2x^3 - 3x^2 + 1$ at the point where $x = 1$. Give your answer in the form $ax + by + c = 0$ with integer coefficients.

Worked solution.

Step 1: Find the y-coordinate.

$$y = 2 - 3 + 1 = 0 \Rightarrow (1, 0)$$

Substitute $x = 1$.

Step 2: Differentiate.

$$\frac{dy}{dx} = 6x^2 - 6x$$

Apply the power rule.

Step 3: Tangent gradient at $x = 1$.

$$6 - 6 = 0$$

The tangent is horizontal.

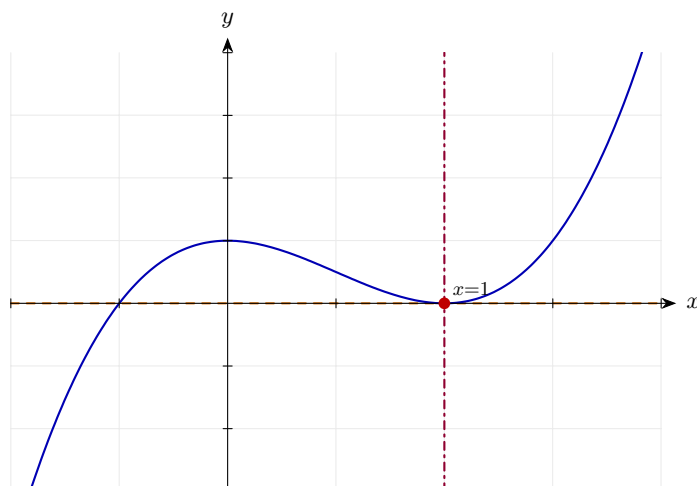
Step 4: Normal is vertical.

$$x = 1 \Leftrightarrow x - 1 = 0$$

A line perpendicular to a horizontal tangent is vertical.

Final answer: $x - 1 = 0$

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 59 (d2-059)

Difficulty: Standard / Marks: 3

Find the coordinates of the stationary point of $y = x^2 - 8x + 5$.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 2x - 8$$

Apply the power rule.

Step 2: Set gradient to zero.

$$2x - 8 = 0 \Rightarrow x = 4$$

Solve for x .

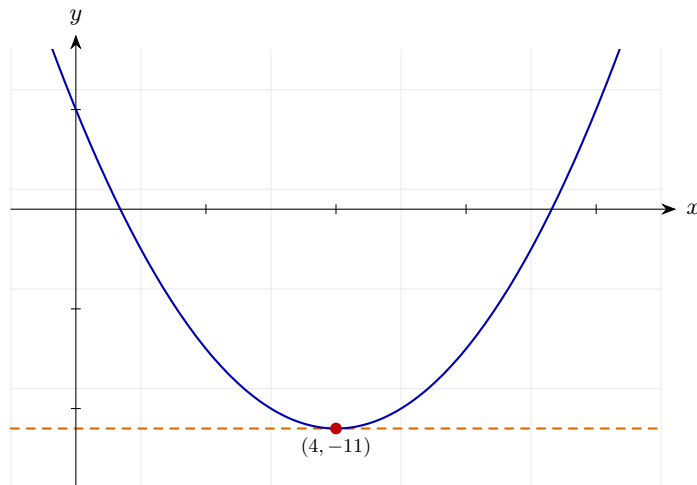
Step 3: Find y .

$$y = 16 - 32 + 5 = -11$$

Substitute.

Final answer: Stationary point at $(4, -11)$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 60 (d2-060)

Difficulty: Standard / Marks: 5

Find the coordinates of the stationary points of $f(x) = x^3 - 3x^2 - 9x + 1$.

Worked solution.

Step 1: Differentiate.

$$f'(x) = 3x^2 - 6x - 9$$

Apply the power rule.

Step 2: Solve $f'(x) = 0$.

$$3x^2 - 6x - 9 = 0 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$$

Factorise.

Step 3: Values of x and corresponding y .

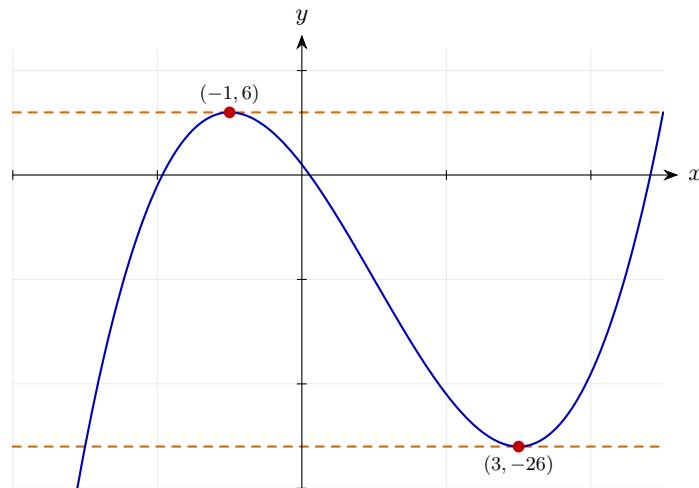
$$x = 3 : f(3) = 27 - 27 - 27 + 1 = -26$$

$$x = -1 : f(-1) = -1 - 3 + 9 + 1 = 6$$

Substitute each value.

Final answer: Stationary points at $(3, -26)$ and $(-1, 6)$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 61 (d2-061)

Difficulty: Standard / Marks: 4

The curve $y = 2x^3 + 3x^2 - 12x + 5$ has two stationary points. Find their x -coordinates.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

Apply the power rule.

Step 2: Set to zero and simplify.

$$6x^2 + 6x - 12 = 0 \Rightarrow x^2 + x - 2 = 0$$

Divide by 6.

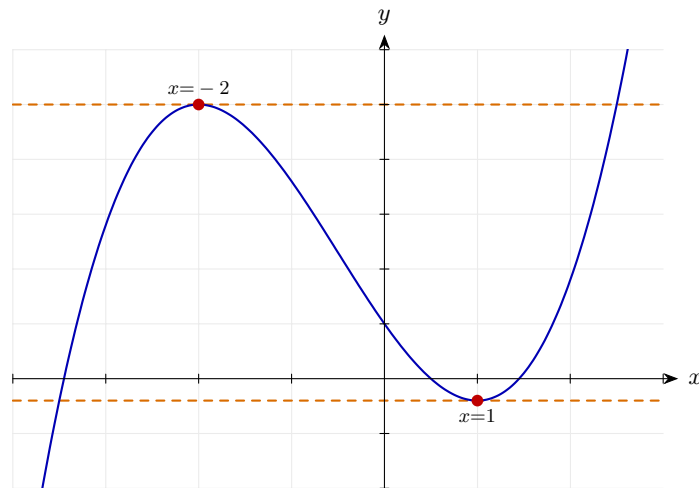
Step 3: Factorise and solve.

$$(x + 2)(x - 1) = 0 \Rightarrow x = -2 \text{ or } x = 1$$

Two stationary points.

Final answer: $x = -2$ and $x = 1$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 62 (d2-062)

Difficulty: Standard / Marks: 5

The curve $y = 4x^3 - 12x + 7$ has a tangent parallel to the x -axis. Find the y -coordinates of the points where this occurs.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 12x^2 - 12$$

Apply the power rule.

Step 2: Tangent parallel to x -axis when gradient = 0.

$$12x^2 - 12 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

Solve for x .

Step 3: Find y at each value.

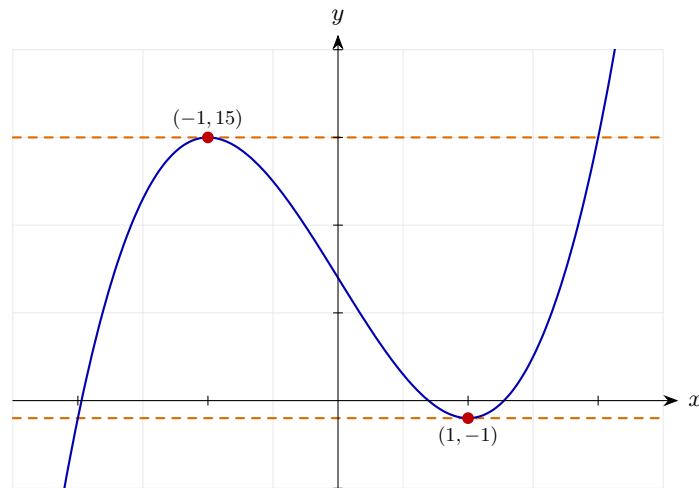
$$x = 1 : 4 - 12 + 7 = -1$$

$$x = -1 : -4 + 12 + 7 = 15$$

Substitute into the original equation.

Final answer: $y = -1$ and $y = 15$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 63 (d2-063)

Difficulty: Standard / Marks: 4

Find the values of x for which the curve $y = x^3 - 3x^2 + 4$ has gradient 9.

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 3x^2 - 6x$$

Apply the power rule.

Step 2: Set equal to 9.

$$3x^2 - 6x = 9 \Rightarrow x^2 - 2x - 3 = 0$$

Rearrange to a quadratic in x .

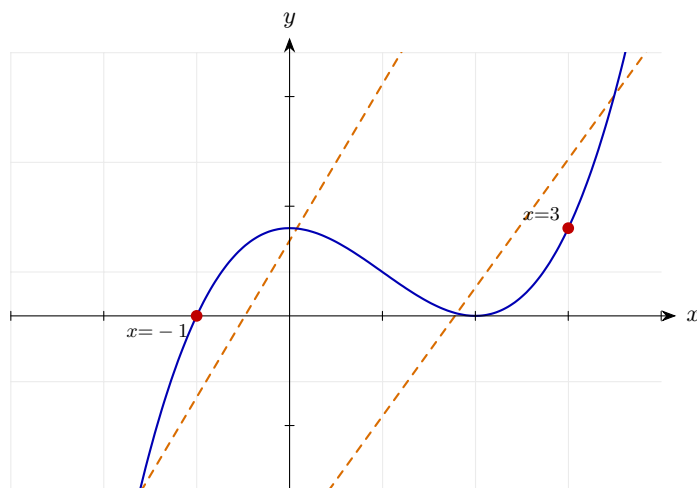
Step 3: Factorise and solve.

$$(x - 3)(x + 1) = 0 \Rightarrow x = 3 \text{ or } x = -1$$

Two solutions.

Final answer: $x = 3$ or $x = -1$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 64 (d2-064)

Difficulty: Standard / Marks: 4

The volume $V \text{ cm}^3$ of a cube of side $x \text{ cm}$ is $V = x^3$. Find $\frac{dV}{dx}$ and evaluate it when $x = 5$. Interpret your answer.

Worked solution.

Step 1: Differentiate.

$$\frac{dV}{dx} = 3x^2$$

Apply the power rule.

Step 2: Substitute $x = 5$.

$$3(25) = 75$$

Value of the derivative.

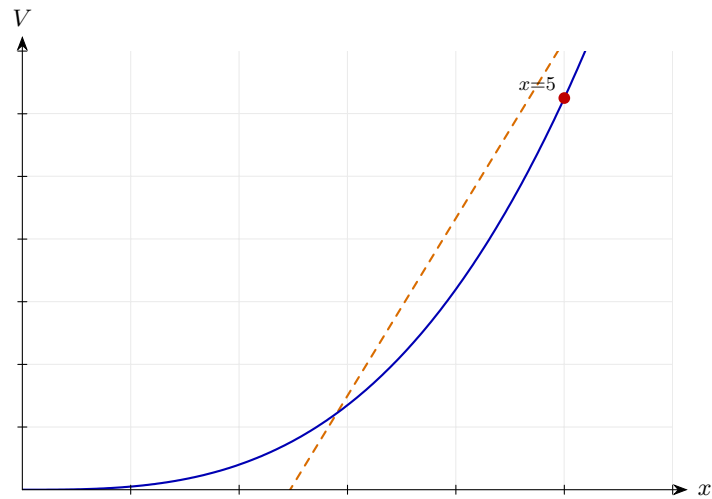
Step 3: Interpret.

$$\left. \frac{dV}{dx} \right|_{x=5} = 75 \text{ cm}^3/\text{cm}$$

When the side is 5 cm, the volume is increasing at 75 cm^3 per unit increase in side length.

Final answer: $\frac{dV}{dx} = 3x^2$; at $x = 5$ it equals 75 cm^3 per cm.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 65 (d2-065)

Difficulty: Standard / Marks: 4

The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. Find the velocity when $t = 2$.

Worked solution.

Step 1: Differentiate to get velocity.

$$v = \frac{ds}{dt} = 3t^2 - 12t + 9$$

Velocity is the derivative of displacement.

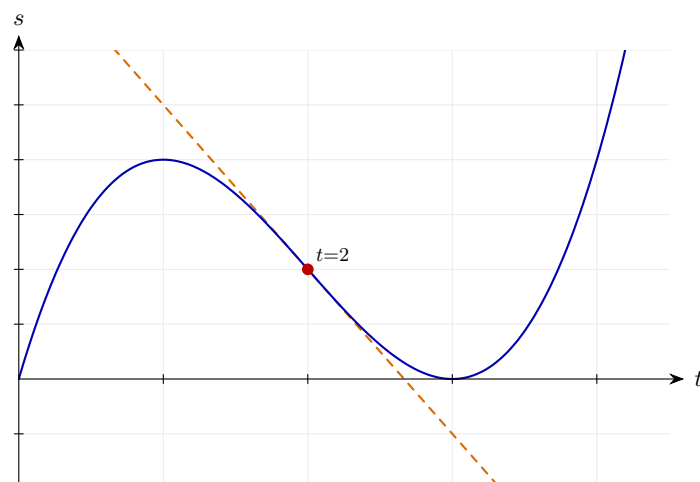
Step 2: Substitute $t = 2$.

$$3(4) - 12(2) + 9 = 12 - 24 + 9 = -3$$

Evaluate.

Final answer: Velocity at $t = 2$ is -3 m/s.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 66 (d2-066)

Difficulty: Standard / Marks: 3

The cost C (in £) of producing x widgets is $C = 200 + 5x + 0.01x^2$. Find the marginal cost $\frac{dC}{dx}$ when $x = 100$.

Worked solution.

Step 1: Differentiate.

$$\frac{dC}{dx} = 5 + 0.02x$$

Constant 200 vanishes.

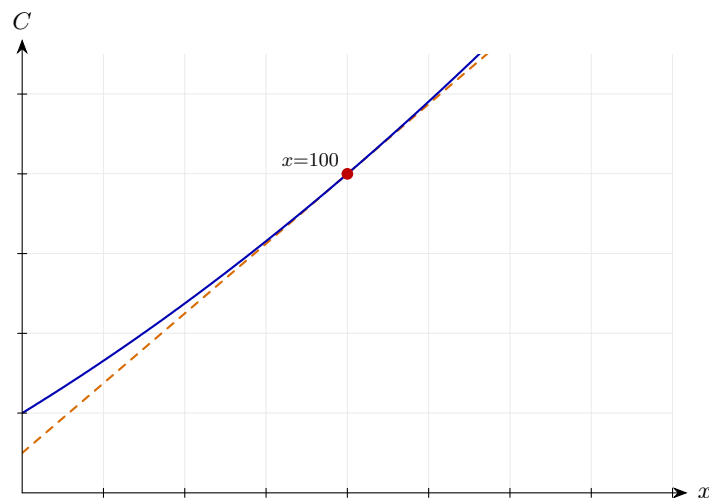
Step 2: Substitute $x = 100$.

$$5 + 2 = 7$$

Marginal cost at 100 widgets.

Final answer: Marginal cost is £7 per widget.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 67 (d2-067)

Difficulty: Standard / Marks: 3

A spherical balloon has volume $V = \frac{4}{3}\pi r^3$ cm³, where r is its radius in cm. Find $\frac{dV}{dr}$ and evaluate it when $r = 10$.

Worked solution.

Step 1: Differentiate with respect to r .

$$\frac{dV}{dr} = \frac{4}{3}\pi \cdot 3r^2 = 4\pi r^2$$

Apply the power rule.

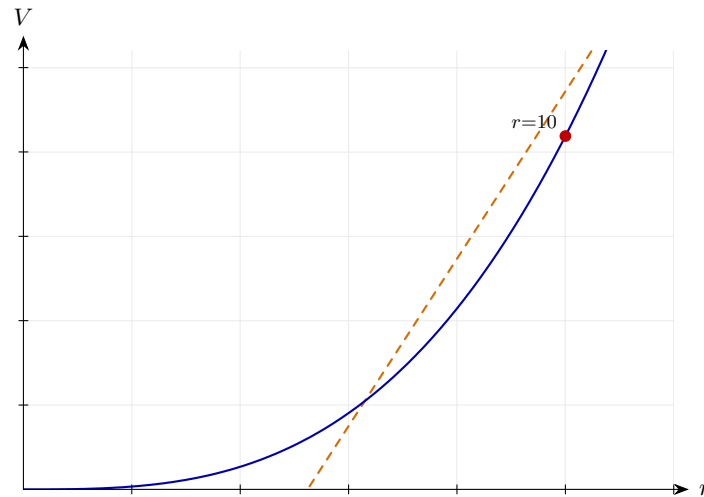
Step 2: Substitute $r = 10$.

$$4\pi(100) = 400\pi$$

Evaluate.

Final answer: $\frac{dV}{dr} = 4\pi r^2$; at $r = 10$ it equals $400\pi \text{ cm}^3$ per cm.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 68 (d2-068)

Difficulty: Challenge / Marks: 6

The curve C has equation $y = x^3 - 6x^2 + 11x - 4$.

(a) Find $\frac{dy}{dx}$.

(b) The tangent at the point P on C is parallel to the line $y = 2x + 1$. Find the x -coordinates of the possible positions of P .

Worked solution.

Step 1: (a) Differentiate.

$$\frac{dy}{dx} = 3x^2 - 12x + 11$$

Apply the power rule.

Step 2: (b) Parallel to $y = 2x + 1$ means gradient is 2.

$$3x^2 - 12x + 11 = 2$$

Set derivative equal to the target gradient.

Step 3: Rearrange.

$$3x^2 - 12x + 9 = 0 \Rightarrow x^2 - 4x + 3 = 0$$

Divide by 3.

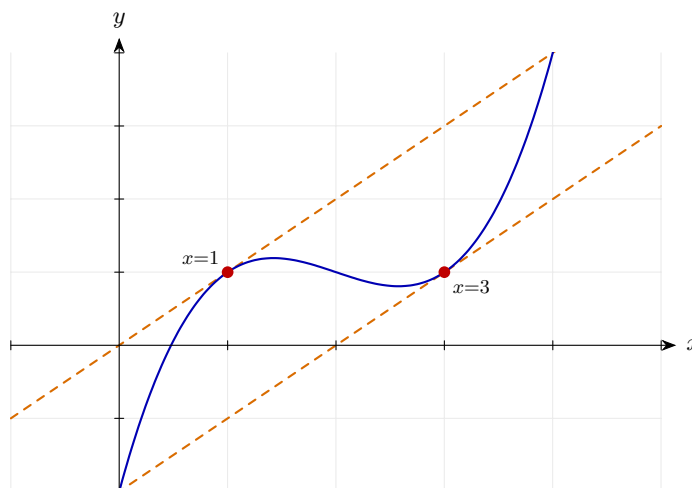
Step 4: Factorise and solve.

$$(x - 1)(x - 3) = 0 \Rightarrow x = 1 \text{ or } x = 3$$

Two possible positions.

Final answer: (a) $\frac{dy}{dx} = 3x^2 - 12x + 11$
 (b) $x = 1$ or $x = 3$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 69 (d2-069)

Difficulty: Challenge / Marks: 7

The curve C has equation $y = 2\sqrt{x} + \frac{8}{\sqrt{x}}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{x - 4}{x\sqrt{x}}$.

(b) Find the coordinates of the stationary point of C .

Worked solution.

Step 1: (a) Write with fractional indices.

$$y = 2x^{1/2} + 8x^{-1/2}$$

Prepare each term for the power rule.

Step 2: Differentiate.

$$\frac{dy}{dx} = x^{-1/2} - 4x^{-3/2}$$

Multiply each term by its index and reduce by 1.

Step 3: Factor out $x^{-3/2}$.

$$= x^{-3/2}(x - 4) = \frac{x - 4}{x\sqrt{x}} \checkmark$$

Matches the required form.

Step 4: (b) Stationary when numerator is zero.

$$x - 4 = 0 \Rightarrow x = 4$$

Denominator non-zero for $x > 0$.

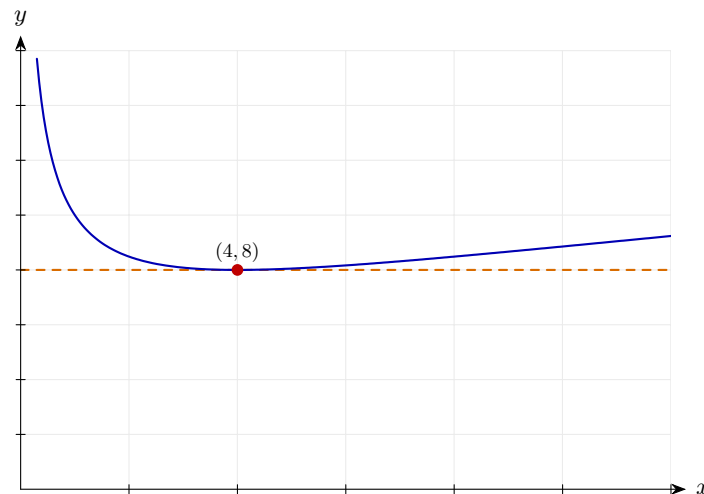
Step 5: Find y .

$$y = 2\sqrt{4} + \frac{8}{\sqrt{4}} = 4 + 4 = 8$$

Substitute back.

Final answer: (a) Shown.
(b) Stationary point at $(4, 8)$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.

Differentiating $y = f(x)$ 70 (d2-070)

Difficulty: Challenge / Marks: 9

The curve C has equation $y = x^3 - 3x^2 - 24x + 1$.

(a) Find the coordinates of the two stationary points of C .

(b) Find the equation of the tangent to C at the point where $x = 0$, giving your answer in the form $y = mx + c$.

(c) Determine the values of x for which C has gradient greater than 0.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{dy}{dx} = 3x^2 - 6x - 24$$

Apply the power rule.

Step 2: Solve gradient = 0.

$$3(x^2 - 2x - 8) = 0 \Rightarrow (x - 4)(x + 2) = 0$$

Factor out 3, then factorise the quadratic.

Step 3: Compute y values.

$$x = 4 : 64 - 48 - 96 + 1 = -79$$

$$x = -2 : -8 - 12 + 48 + 1 = 29$$

Substitute into the original equation.

Step 4: (b) Tangent at $x = 0$.

$$y(0) = 1, \quad \frac{dy}{dx}(0) = -24$$

Evaluate the curve and gradient at 0.

Step 5: Tangent equation.

$$y - 1 = -24(x - 0) \Rightarrow y = -24x + 1$$

Point-slope form.

Step 6: (c) Curve increasing where derivative > 0 .

$$(x - 4)(x + 2) > 0 \Rightarrow x < -2 \text{ or } x > 4$$

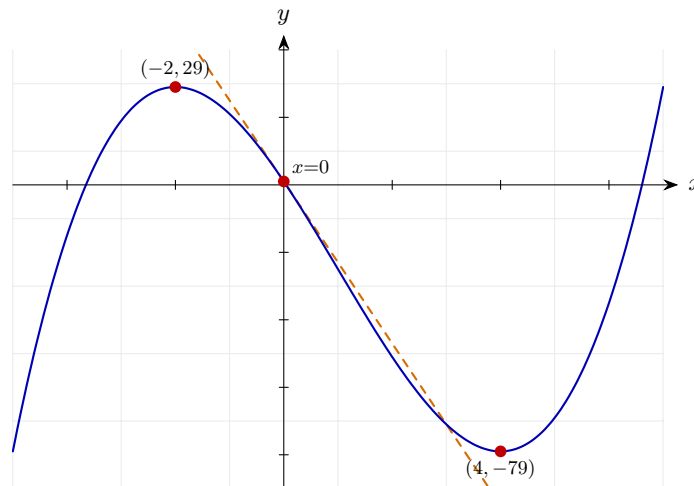
Quadratic positive outside its roots.

Final answer: (a) $(-2, 29)$ and $(4, -79)$

(b) $y = -24x + 1$

(c) $x < -2$ or $x > 4$.

Diagram.



Curve (blue); tangent (dashed orange); normal (dash-dotted purple) / derivative (teal) as relevant.
